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RISK ANALYSIS AND SAFETY REVIEW PACK

Document: RISK-EEG-011 **Revision:** B **Date:** 2026-09-01 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0
Governing documents: DSN-EEG-003 Rev D, then RFQ-EEG-001 Rev E. Where this document and design.py disagree, design.py governs.

Why this document exists

RFQ-EEG-001 section 9.2 makes an electrical safety review the gate between two prototypes and ten kits worn by human participants, and DSN-EEG-003 Rev D section 8 records that no safety engineer has reviewed any of this design. Until now that gate had no input document and no form: there was no hazard analysis, no patient auxiliary current calculation, no isolation report, no checklist and nothing to sign. This pack is that input. It analyses the instrument as an ISO 14971-style risk file, works the S-02 current budget through with the real parts of EEG-CAR-01 Rev B, states what the isolation barrier is and where it lives, sets out battery safety for a kit that travels by post, and ends with the checklist, acceptance criteria and signature block the reviewer works to. It is written on the assumption the reviewer is a stranger to the programme and hostile to hand-waving. **Nothing in this package has been built, and no safety engineer has reviewed this design.** Nothing in it has been measured. Every figure marked *calculated* is exactly that, and the review's first job is to decide whether a calculation is enough evidence for the decision it supports.

What changed in Rev B. Two things about the carrier changed during layout and they change parts of this pack. The board grew from 130 × 124 mm to **150.0 × 130.0 mm**, because thirty connectors, 211 parts and 156 nets would not close at the smaller size. The board went from two layers to **four** – L1 signal, L2 reference plane, L3 reference plane, L4 signal – because a two-layer carrier cannot be both a continuous reference under every analogue trace and a second routing surface at the same time. Package v1 asserted that two layers would be enough; doing the layout showed that it is not. Section 5 explains what four layers means for the isolation argument, and it is more work to verify, not less: the keep-out under the barrier must now be proved on four copper layers, two of which are invisible from the outside of a finished board. Section 5.5 records what the design rule check now says about both halves of that: the isolation rule area is clean on all four layers, and the report now records zero violations of any rule, with all 145 nets connected. The fabrication data is **released for review** under RFQ-EEG-002A. It is not released for fabrication: the routing was produced by the programme's own tools and no human layout engineer has read it.

Corrections within Rev B (2026-09-01). The findings of the second cross-document audit that name this pack are closed in this issue, within the same release and without a revision letter change. They were the DRC state and its place as a release gate (section 5.5, SR-11, checklist item 4.16), the isolation-strip claim, which is now made because the DRC report records it rather than because a rule area was drawn, and E-10, whose two states are now stated together everywhere the requirement appears. The binding answers are in **RUL-EEG-021 Rev B**, the rulings register, sections A, B and C; where this pack and tools/design.py disagree, design.py governs.

Corrections within Rev B (2026-09-02), after the design changes of 2 September 2026. Four changes were made in the source today and all four reach this pack, so each is written up where the pack already analyses the thing it changes. **Channel 11 is no longer an exception.** The override in `tools/design.py` that put D11.3 and C11.1 on BIAS_EL, the patient side of R11, is deleted; the row is now generated by the same loop as the other fifteen, so the clamp and the filter sit on BIASOUT, behind the series protection resistor, like the other fifteen. SF-1a and SF-6a of section 4.4 are closed by that change and are restated below with the ordinary SF-1 figures. Those figures moved again the same day, when ECO-EEG-024 was applied: they are **36.8 μ A bound A and 30.0 μ A bound B** at the 68 k Ω now fitted, and the 53.2 μ A / 41.2 μ A this paragraph carried at first issue are superseded from 2026-09-02 (see the ECO-EEG-024 paragraph below). The 183.6 μ A that SF-1a carried is withdrawn. **SR-12 is closed in the design and its disposition is still owed**, because applying the fix an analysis points to is not the same as having the reviewer who owns the finding approve it. **The HM-04 contact-light window is now two pockets** with 1.60 mm of printed PA12 between them – an outboard LED seat and an inboard conductor run – where it was one box straight through the body. That materially reduces SF-9 and strengthens the basis of H-05, and it does not eliminate either: nothing has been printed, and SF-9's own control is physical separation *and its verification*, of which only the separation half has moved. **HM-04's spring seat is deepened** from 4.50 to 6.60 mm, which takes the free height above the carrier's spigot from 1.40 mm to 3.50 mm; that is written up at H-17, whose control is a 3 –6 N spring that could not have been bought in the volume the released geometry left for it and that is still unspecified. **E-27's bicolour phase driver is written**, so the sentence H-25 carried, that the driver was specified and not yet coded, is superseded on this date. Nothing in this package has been built or measured; the mechanical figures quoted below and in PARTS-EEG-019 are booleans and dimensions taken from solid models, not from parts. One consequence reached the routing, and it has since been worked through. **Corrected 2026-09-02:** at first issue this paragraph said the carrier *was being re-routed* for the channel 11 change, because moving D11.3 and C11.1 swaps R11's pads and changes the BIAS_EL and BIASOUT nets. That re-route is done. `kicad/EEG-CAR-01_RevB_DRC_report.txt` of 2026-09-02 reports **145 of 145 nets fully connected, VIOLATIONS: 0, 3 745 track segments and 552 vias**, with the isolation rule area clean on all four copper layers and the 0.35 mm electrode clearance met, so every statement in section 5.5, in section 7 and in SR-11 that rests on the design-rule report now rests on the post-change report and not on a superseded one. Two things are recorded rather than assumed. The router still had to relax **169 connections** to close, which is the figure those sections already carry. And the relaxed-geometry connection the report lists twice as "BIASOUT J2.10 -> R11.1", at 0.20 mm track and 0.35 mm gap, was **expected to retire with the change and did not**: it is still listed, at the same geometry, in the post-change report. The expectation is withdrawn; the connection is one of the 169 the RFQ-EEG-002A reviewer is asked to look at.

Corrections within Rev B (2026-09-02), after ECO-EEG-024 was applied. On the same day, and after the bias-topology change above, the programme took **disposition 2 of section 4.5:** `tools/design.py:88` now fits **R1–R16 = 68 k Ω 0.1 % 25 ppm thin film** (Vishay TNPW060368K0BEEA) in place of 47 k Ω , in the same 0603 footprint, on the same nets, so sixteen parts change and the board does not move. Every single-fault figure in section 4 moves with them: **bound A from 53.2 μ A to 36.8 μ A, bound B from 41.2 μ A to 30.0 μ A**, against S-02's 50 μ A limit. **S-02 is therefore met in the design from 2026-09-02**, on bound A, which is the bound this pack judges against and the worse of the two. The 53.2 μ A and 41.2 μ A figures are superseded as statements of the live design and are kept below only where this pack records what it corrected. E-10 moves with the resistors onto its **± 1.0 dB** branch, which the requirement already states

and which is earned by exactly this change and by nothing else; TST-EEG-004 T22 grades against it.

SR-01 is not closed by this, and this pack does not close it. The finding belongs to the electrical safety reviewer of section 7, the review has not started, nothing in this package has been built or measured, and a programme applying the fix its own analysis recommended is not the same as the reviewer who owns the finding accepting it. What the change alters is the question the reviewer is asked: they are now handed a design that **meets** the limit by calculation and asked to confirm it, rather than one that does not and asked to accept it. Section 10 keeps SR-01 open, and section 7.3 item 4.3 is the form it is disposed of on.

1. Intended use, users, use environment

1.1 Intended use

The EEG field kit is a sixteen-channel research instrument that records scalp EEG, facial and laryngeal EMG, and three audio envelope channels, together with button responses and audio, during a scripted listening and speaking protocol. It streams that data over an isolated USB link to a browser-based session runner and writes an identical copy to a microSD card.

It exists to produce group-level data for one pre-registered study. It performs no analysis on the device, applies no algorithm to a participant's data in real time, and returns no result of any kind to the person wearing it.

1.2 This is not a medical device

The instrument has no diagnostic, therapeutic, monitoring, prognostic or screening purpose. It is not placed on the market. It is loaned, not sold. It is not a medical device within Regulation (EU) 2017/745, and the programme makes no claim that it is one. REG-EEG-012 carries the declaration, the reasoning and the controlled wording; M-03 puts the sentence “research instrument – not a medical device” on every enclosure label.

The design nonetheless follows the applicable principles of IEC 60601-1 for a type BF applied part. Those two statements are not in tension and the reviewer should not read them as such: the programme designs to a medical-electrical standard because the device touches a person for two hours at a time, and claims no certification because it has no medical purpose and is not being sold. Where this pack cites IEC 60601-1 it cites it as a design reference, not as a conformity claim.

The word *patient* appears in this document only inside the standard's own term “patient auxiliary current”. Everywhere else the person wearing the instrument is a **participant**.

1.3 Users

User	Who they are	What they do	Training
Participant	An adult who has consented to the study and reports voice-to-skull experiences. Not a clinician. Alone in their own home	Unpacks the kit, applies paste, puts the helmet on, follows the on-screen runner, presses buttons, ends the session, repacks and posts the kit back	A laminated card (IFU-EEG-014), an on-screen guide, and a support telephone number. No supervision, no site visit
Operator	Programme staff at Brussels	Refurbishes returning kits, replaces consumables and electrode cups with the HM-09 service key, runs the incoming test subset, despatches	Trained on SVC-EEG-013
Manufacturer	The contract manufacturer	Builds, provisions, tests to TST-EEG-004 Rev C, packs	IPC-A-610 class 2, plus this pack for the risk-control components
Safety reviewer	An external engineer engaged by the programme	Section 7 of this document	See 7.1

1.4 Use environment

A domestic room. Uncontrolled temperature (assume 15 to 35 °C), uncontrolled humidity, mains sockets of unknown quality, a mains-powered laptop of unknown make on the same desk, pets, children, furniture, and no one else present. The kit travels between uses by courier in a hard case, is handled by carriers who do not know what is inside, and is unpacked by someone who has never seen it before.

Sessions are two to three per participant, each about two hours, of which up to two hours is with the helmet on the head and headphones on the ears. A kit serves about twenty participants a year in sequence, and more than that over the fleet's life; the arithmetic is SVC-EEG-013 section 2, which costs fourteen days with a participant and about twenty turnarounds per kit per year. The occurrence scores in section 3 are written against the annual figure, and a multi-year fleet raises the exposed population above it.

1.5 Reasonably foreseeable misuse

Charging while wearing the helmet. Opening the battery hatch. Wearing it while cooking, driving, or lying down. Wearing it overnight. Substituting other headphones. Letting a child or another adult wear it. Immersing the helmet in water to clean it. Using the recording as evidence of an external attack, and being distressed when told it is not.

2. Hazard identification

Hazards are grouped by energy or agent. Each group ends with the design property that is supposed to hold it, so the reviewer can attack the property rather than the prose.

Electrical. The only mains-referenced conductor anywhere in the system is on the host side of the ADuM4160 module at J10. The carrier has no host-side copper at all: J10's four pins are VDD_ISO, USB_DN, USB_DP and DGND, all device-side (verified against design.py). The host connection is a socket and not a captive cable – the isolator module's own host receptacle, USB-B on the only named candidate and brought to a panel USB-C by the WH-09 pigtail, which is a live non-conformance against E-24 (section 5.4), presented through a gasketed aperture in POD-P1 – so exactly one externally accessible data connector exists on the pod. Everything on the participant's side of the barrier is a floating island powered by one 18650 cell. The credible electrical hazards are therefore (a) failure of that barrier with a mains-referenced host attached; (b) a conductive path that bypasses the barrier altogether – specifically the ESP32-S3-DevKitC-1's own two USB-C receptacles, which sit on GPIO19/20, the same pins the carrier taps for J10, and are on the non-isolated side; (c) a fault putting an internal rail on an electrode; (d) the charge input at J24, which is fed from a consumer mains adaptor whose isolation the programme does not control.

Thermal. About 0.6 W dissipated inside a printed enclosure worn against the occiput, and about 1.2 W while charging. Cell abuse is covered under battery.

Mechanical. A 1 m drop of a packed case. The chin strap. The 1.2 m headphone cable. The printed bayonets and the frame edges. Pressure from eight spring-loaded electrode assemblies over two hours. A crushed cell after a drop.

Chemical. Conductive paste and abrasive prep gel applied by an untrained person to their own scalp, near their own eyes. 70 % IPA, which is an operator agent at refurbishment and is deliberately not in the participant kit. Biocompatibility of the skin-contact materials listed in REG-EEG-012 §6.2, none of which currently has a declaration on file. That list is eight rows, this pack counts the same eight, and SR-05 is written against every row on it.

Biological. About twenty people a year, and more over the fleet's life, wear the same frame, the same ear clips, the same boom and the same ear cushions in sequence, with paste and abraded skin in the loop.

Information. A microSD card holding raw EEG, the participant's voice and room audio travels back through the post. A unit can be confused with another unit, or its calibration record attached to the wrong serial, which corrupts data rather than injuring anyone – but the study is the reason the device exists, so a data hazard is a real hazard.

Psychological. This cohort has been recruited because they report being acted on and controlled by others at a distance. A strap under the chin, eight lights on the head, a microphone pointing at the room and a device described as “recording your brain” are not neutral objects to them. A red light, a failed session, a device that will not connect, or an instrument that appears to switch itself on can be incorporated into a persecutory belief, at home, at night, with no clinician within reach. This is not a soft category appended for completeness; on this programme it is the hazard with the highest expected frequency.

3. Risk table

Severity. S1 negligible (no injury, momentary discomfort) · S2 minor (reversible, no intervention) · S3 serious (reversible, needs intervention) · S4 critical (irreversible impairment) · S5 catastrophic (death).

Probability, per participant-session. The whole programme is about 3,000 participant-sessions at 50 kits, so P1 means “not expected to happen once in the programme”. P1 improbable ($< 1e-5$) · P2 remote ($1e-4$ to $1e-5$) · P3 occasional ($1e-3$ to $1e-4$) · P4 probable ($1e-2$ to $1e-3$) · P5 frequent ($> 1e-2$).

Risk index $R = S \times P$. **1–4** acceptable · **5–9** acceptable with the control in place and the residual stated to the participant · **10–14** tolerable only on the reviewer’s written acceptance · **15–25** unacceptable, must be reduced before use on a person.

Probabilities are engineering estimates by the programme, not field data. There is no field data. The reviewer may reject any of them.

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-01	Mains energy via the host	Laptop on mains → ADuM4160 barrier fails → host earth reaches the floating island → current through the participant	Burn, ventricular fibrillation	5	1	5	ADuM4160 module, 5 kV RMS reinforced part (S-03, E-24); no carrier copper crosses the barrier; the isolation keep-out of DSN-EEG-003 Rev D §3.3 rule 4, enforced on all four copper layers; R1–R16 limit any resulting current	T20 per unit (500 V DC, $\geq 1 \text{ G}\Omega$); supplier isolation certificate at T00; artwork creepage check on all four layer plots once per Gerber revision (TST-EEG-004 section 14). The Rev B DRC report lists no isolation keep-out violation on any of the four layers, and no violation of any other rule either; the check is still repeated on the Gerbers finally released, because no layout engineer has yet read this routing – see 5.5 and SR-11	5 / 1 / 5
H-02	Barrier bypassed at the DevKit	Participant or operator plugs a mains-powered laptop into one of the DevKitC-1’s own USB-C sockets instead of the isolator’s	Same as H-01, with the barrier absent	5	3	15	Both DevKitC-1 receptacles physically blanked by the POD-P1 wall and the HM-01 shell (ASM-EEG-007 final check); one externally accessible data connector only, the isolator module’s own USB-C through the POD-P1 aperture	Final-assembly safety check before labelling; first-article inspection; photograph in the unit record	5 / 1 / 5

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-03	Excessive auxiliary current, single fault	One BAV99 half fails short → AVDD or AVSS on the module-side node → current out through the series resistor into the scalp	Sensation, in the limit a burn at a small contact	3	2	6	R1–R16 68 kΩ 0.1 % (ECO-EEG-024, applied 2026-09-02; 47 kΩ until that date); ±2.5 V rails rather than 5 V; zone split at x = 62 mm keeps V5V and DVDD3V3 away from every electrode net, on the signal layers and in the inner planes	Section 4 calculation; T23 per unit. The calculated bare-resistor bound is 36.8 μA against a 50 μA limit, a 26 % margin, so S-02 is met in the design. <i>Corrected 2026-09-02: this cell read “53.2 μA against a 50 μA limit, so S-02 is not met” while 47 kΩ was fitted.</i> The row is still not closed , for two reasons that the resistor change does not touch: nothing has been measured, and SR-01 is the safety reviewer's finding and is open for their written disposition	3 / 2 / 6, not closed – met in the design, open for sign-off and unmeasured
H-04	Excessive auxiliary current, normal	Clamp and input leakage sum at the participant across 14 connections at 60 °C	Sensation	2	1	2	Same components; C0G filter capacitors; AC lead-off at 6 nA (E-06)	Section 4: 3.51 μA calculated against 10 μA; T23 per unit	2 / 1 / 2
H-05	Light drive into an electrode	An LED conductor contacts an electrode conductor inside HM-01	Sensation, tissue damage at 3.3 mA	3	1	3	ECO-EEG-014: two physically separate cables, WH-01 screened 12-way to J14 and WH-02 10-way to J30, separate glands, separate internal channels. Added 2026-09-02: the two conductors are also separated at the eight electrodes themselves, by 1.60 mm of printed PA12 between an outboard LED pocket and an inboard conductor pocket in hm04 () , where the released body carried one undivided window through it – see SF-9	Harness insulation resistance ≥ 100 MΩ at 100 V DC between WH-01 and WH-02 at acceptance (WH-EEG-008); routing inspected at helmet build. The site half of this control is drawn and not verified: no HM-04 has been printed, the LED seat and its lead passages are still a WH-EEG-008 §3.1.1 proposal, and no insulation figure across the 1.60 mm web exists. The probability stays at P1 because the two-cable architecture already carried it; what changed is that the estimate is no longer contradicted by the electrode body, which is what ASM-EEG-007 open item 20 recorded	3 / 1 / 3
H-06	Screen short in the electrode cable	One electrode conductor shorts to the drain wire inside WH-01	Sensation	3	2	6	The 68 kΩ series resistor is still in circuit; PTFE insulation tested ≥ 100 MΩ at 500 V DC; HARN_SHIELD joins DGND at R91 only	Section 4 SF-3: 36.8 μA calculated, inside the 50 μA limit. <i>Corrected 2026-09-02: this cell read “WH-EEG-008 §5.5: 53 μA calculated, over the limit” against the 47 kΩ then fitted.</i> The same open point as SR-01 – nothing is measured and the reviewer's disposition is owed	3 / 2 / 6, not closed – met in the design, open for sign-off and unmeasured

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-07	Charging while worn	Participant plugs the charger in mid-session; both firmware interlocks fail	Participant island referenced to a consumer adaptor's secondary; up to ~250 μ A AC	3	2	6	S-01 two mechanisms: session start refused while VBUS_DET is high, and CHG_CE held asserted for the session; separate charge-only receptacle at J24; label and participant card say it	T4 and T21 per unit	3 / 2 / 6
H-08	Both interlocks are firmware	A firmware defect defeats S-01; there is no independent hardware inhibit	As H-07	3	2	6	Two independent code paths, one sensing hardware (R84/R85), one driving hardware (CHG_CE)	T21 exercises both; no hardware AND exists – see SR-02	3 / 2 / 6, decision owed
H-09	Cell thermal event in use	Internal short, crush after a drop, or charger fault → venting or fire on or near the head	Burn, inhalation injury	5	1	5	Protected 18650 with PCM; bq24074-class charger with thermal regulation; PTC F1 (1.1 A hold / 2.2 A trip) and TVS D24 on the charge input; polarised JST PH at J13 and the HM-10 keyed cell carrier	T00 cell lot documents (UN 38.3, IEC 62133-2) and cell OCV; M-04 drop with cell inspection	5 / 1 / 5
H-10	Charging outside the safe temperature window	Kit charged in a cold or hot room; no thermistor path exists on the carrier	Lithium plating, later thermal event	4	3	12	bq24074 internal thermal regulation only; participant instruction to charge at room temperature	J12 has no TS pin and J13 is a 2-pin connector, so S-04's thermistor is not fitted and E-23's 45 °C inhibit is untested and untestable. Neither requirement is met – see SR-03	4 / 3 / 12, not closed
H-11	Cell damaged in the return post	Participant drops the case; kit returns with a deformed or swollen cell	Fire in a courier vehicle or at goods-in	5	2	10	State of charge $\leq$ 30 % at despatch (S-09); cell inside equipment, retained, not loose; hard case; UN3481 PI967 section II marking and carrier documentation	Goods-in quarantine procedure (section 6.5); SVC-EEG-013 damage schedule	5 / 1 / 5
H-12	Surface too hot against the skin	Two-hour session, pod at the occiput, warm room	Erythema, discomfort	2	2	4	About 0.6 W total; calculated rise 0.9 K in POD-P1 and 1.9 K in the smaller Phase 2 occipital shell; charging inhibited during a session	3 h thermal type test with thermocouples on the TPU pads; limit 43 °C at any skin-contact surface	2 / 1 / 2

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-13	Acoustic overexposure	Register default, corrupt stimulus file or calibration error drives the ES8388 headphone amplifier to full scale	Temporary threshold shift; in the limit permanent hearing damage	4	3	12	RFQ E-29: no more than 100 dB SPL at any commanded level, with the codec volume register clamped in firmware to the value measured at calibration; the clamp and level registers read back per unit at TST-EEG-004 T17 and T6 , not at run time – the F-09 STATUS frame carries no level field; SW3 stop button; E-16 tap taken before any adjustable control	E-29 type test on an artificial ear, TST-EEG-004 T28 , listed in TST section 14, once per lot; T17 headphone level per unit. The ceiling is firmware only, there is no hardware limit, and calculated full-scale output is about 110 dB SPL – see SR-04	4 / 2 / 8, not closed
H-14	Chin strap entrapment	Strap catches on furniture or a chair arm as the participant stands	Neck compression, fall	3	2	6	HM-06 chin strap is removable and the runner says so before it is first mentioned; two anchor points at ear level so no loop can form around the neck; side-release buckle openable one-handed	Fit verification on a headform; the removability statement checked in the participant card	2 / 2 / 4
H-15	Cable entanglement	1.2 m headphone lead and the helmet harness while moving or reclining	Fall, device pulled from the head	2	3	6	Single lead, no loops; instruction not to wear while lying down, sleeping, cooking or driving; kit used seated at a desk	Participant card content review	2 / 2 / 4
H-16	Drop damage in transit	Courier drops the case	Cracked frame, broken conductor, crushed cell	2	4	8	IP67 hard case; the CASE-00 Rev C die-cut PE insert of PKG-EEG-015 section 2.2 – seven loose-laid 25 mm layers, 175 mm of stack, nine bays, every bay at least 3 mm larger than its part with 8 mm of foam between bays, the helmet hung at the halo on a shelf in the top layer with the electrode cups free of every surface; labelled bays and a packing photograph in the lid; outer carton (M-07). The two-sheet insert this row named at the first issue of this revision is withdrawn: PKG-EEG-015 section 2.4 shows 50 mm of foam cannot hold a 158 mm helmet	M-04 1 m drop type test on each face, run on the Rev C insert and not on the superseded two-sheet one, followed by a re-run of T3, T7, T8, T10, T14 to the same limits. The seven Rev C cut files are drawn and shipped – mech/CASE-00_foam_layer_1.dxf to _7.dxf , the Rev B pair deleted – and they are not released for cutting until the bought shell is measured (PKG-EEG-015 section 3.2), so no insert has been cut : the type test still has nothing to run against and this row's control is a schedule, not an insert	2 / 3 / 6

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-17	Pressure injury	Eight assemblies at 3–6 N spring load for two hours	Local erythema, in the limit pressure necrosis	2	3	6	Contact force set by the spring, deliberately independent of the ratchet dial and the strap, so tightening cannot increase it; TPU 85A pads spread frame load	Fit verification; participant card instruction to stop if a site becomes painful. The spring this control rests on is not yet a part. AVL-EEG-017 K12 reads “3–6 N at working length, stainless 302”, selected against the outside of the HM-04 body rather than against the seat the spring sits in, and it has never been specified against a geometry that could hold it: until 2026-09-02 the free height above the HM-05B spigot was 1.40 mm in an annulus 0.10 mm wide on the radius. HM-04's seat was deepened that day and the free height is now 3.50 mm , so a 3–6 N spring is at least realisable, and it is still unspecified and unmeasured (PARTS-EEG-019 OA-20; ASM-EEG-023 Rev B, D5-K12-SPRING-ENVELOPE). Two things go to the safety reviewer with it: the force itself is the one thing the participant cannot adjust, so it is measured on five samples before any fleet order; and the spring's solid height must stay below its working height through the whole 0.40 mm of travel , because a stack that goes solid inside travel stops being a spring and transmits helmet force straight into the scalp for a two-hour session	2 / 2 / 4
H-18	Abrasive prep gel injury	Untrained participant abrades their own scalp too hard, or gets gel in an eye	Skin break, eye irritation	2	3	6	Quantity limited to one 100 g tube; blunt-tip syringes; SDS in the kit; explicit instruction and a contraindication list; no IPA in the participant kit	IFU-EEG-014 content review; SDS on file per lot	2 / 2 / 4
H-19	Skin sensitisation	Two hours' contact with dyed MJF PA12, TPU 85A, paste and ear cushions, repeated	Irritant or allergic contact dermatitis	2	3	6	ISO 10993-5 and -10 declarations required from every supplier (S-05); TPU pads and chin liner replaced every turnaround	No declaration is on file for any material row in REG-EEG-012 §6.2, so S-05 is not met – see SR-05; REG-EEG-012 §6 defines a valid declaration	2 / 3 / 6, not closed

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-20	Cross-participant infection	Frame, ear clips, boom, cups and ear cushions circulate through about twenty people a year, and more over the fleet's life, with paste and abraded skin in the loop	Transmissible skin or scalp infection	3	3	9	Participant touches nothing removable; cups released with the HM-09 service key at refurbishment, ultrasonic cleaned and disinfected; TPU pads, chin liner and EMG pads discarded and replaced; frame wiped with 70 % IPA and never immersed	SVC-EEG-013 procedure with named agent and contact time; refurbishment record per kit. The occurrence score is set against the annual population of about twenty and has not been re-scored against the fleet-life total; that is done at the safety review, section 7.3 item 4.14	3 / 2 / 6
H-21	Reprocessing degrades the parts	Bayonet seizes or the dye leaches after repeated IPA and ultrasonic cycles	A kit cannot be reprocessed; a declaration no longer holds	2	4	8	25-cycle release-and-refit validation with disinfectant exposure on three assemblies, measuring release torque and retention at every cycle	DSN-EEG-002 finding 8 is recorded as not resolved ; the validation belongs in the Phase 1 test plan – see SR-07	2 / 3 / 6, not closed
H-22	Data loss in the post	Case lost or opened in transit with the microSD card in it	Disclosure of raw EEG, the participant's voice and their home audio	3	3	9	Card holds pseudonymous kit and session identifiers only, never a name; card secure-erased at refurbishment after ingest; signed block chain detects tampering; the DPIA states the residual	REG-EEG-012 DPIA section; refurbishment record	3 / 3 / 9
H-23	Unit mis-identification	A calibration record is attached to the wrong serial, or a module is swapped after provisioning	Wrong constants applied to a participant's data; a silent confound across the study	3	3	9	ATECC608B P-256 key is the device identity; iSerial, label and calibration record all derive from it in the TIOV - B-nnnn form defined in PKG-EEG-015 section 5; server verifies every signed block against the exported public key	T5, T6, T16 and the T18 label check; QP-EEG-010 forbids module swaps without full re-characterisation	3 / 2 / 6

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-24	Contact light misreports	Clamp leakage or a lead-off calibration error puts a green light on a poor contact, or a red light on a good one	Unusable data; or, in this cohort, distress at a device that "will not work"	2	4	8	Lead-off is AC at 7.8 or 31.2 Hz and is detected at that frequency, so DC clamp leakage does not enter the reading; per-unit lead-off calibration against fitted references	T10 calibration against the three FIX-01/A 0.1 % thin-film references 4k99, 10k0 and 49k9 , recording the OPEN state and each reference on all eight channels and fitting a per-channel offset and slope by least squares (TST-EEG-004 section 14 T10; "5 k / 10 k / 50 k" was this row's rounding of those E96 parts, and it mattered once – 49k9 through 47 kΩ and the 10 nF shunt read 92.5 kΩ, not the 92.6 kΩ that a true 50 k would give); T11 mapping. Restated 2026-09-02 for ECO-EEG-024's 68 kΩ: TST-EEG-004 T10 gives the three raw expectations as 70.5, 75.1 and 111.5 kΩ at 7.8 Hz where they were 50.7, 55.5 and 92.5 kΩ at 47 kΩ, and the constants are re-derived from them; the 92.5 kΩ figure survives only as the arithmetic that settled the E96 naming. Calculated DC offset from 250 nA through 68 kΩ is 17.0 mV, 9.1 % of the ±187.5 mV input range at gain 24, where it was 11.8 mV and 6.3 % at 47 kΩ (section 4.5)	2 / 3 / 6

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-25	Device appears to act on its own	Contact lights come up at power-on, or during a recording block	Reinforcement of a persecutory belief, acute distress, at home, alone	3	2	6	LED_V is GPIO48, an input at reset: nothing can light at boot whatever the shift register holds. Lights forced dark during every recording block (E-27). R87 ties OE active and R88/C88 give the register a defined power-on state	T11 verifies both the mapping and the forced-dark state; the boot behaviour is a designed property of the pin, checked at first article. The bicolour phase driver was written on 2026-09-02 (firmware/main/main . C), so the sentence this row carried until then – that the driver was specified and not yet coded and that T11 could not pass – is superseded on that date. Both LOFF_STAT P and LOFF_STAT N are now captured, which is what three colours need and one status byte cannot give: neither detector gone is green, exactly one gone is amber, both gone is red. Green is lit in phase A only, red in phase B only and amber in both. LIGHT_PHASE_HZ is 240, a 2.083 ms half-phase, but the alternation is a task delay and quantises to the FreeRTOS tick: 2 ms half-phases on a 1 kHz tick, so the rate on the built image is about 250 Hz, not exactly 240. That meets the “above 100 Hz” this control is written against, the two half-phases stay equal so the duty is 50/50 and the colour does not shift, and it is stated here rather than rounded away. LED_V is returned to an input whenever the lights are dark, which is the state the pin holds at reset and is the reason nothing can light at boot. T11 can now be run and has not been run: no unit exists, nothing has been measured, and the boot behaviour is still a designed property of the pin	3 / 1 / 3

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-26	Radio suspicion	The ESP32-S3 contains a radio; the participant knows or discovers this	Belief that the instrument transmits; withdrawal; distress	3	4	12	Radio never initialised, no credentials stored (E-18, F-19); the fact is stated plainly to the participant together with how it is checked	T24 per unit sweeps 2.400–2.4835 GHz for 60 s during a recording block; result and firmware hash in the unit record	3 / 2 / 6
H-27	Room microphone suspicion	An outward-facing microphone in the participant's home	Belief that the home is being monitored; withdrawal	3	4	12	Hardware mute on MIC_MUTE (GPIO21) reaching both J28.4 and J9.4; audio captured only in scripted windows; the mute state is shown in the runner and logged	T17 verifies the mute; T00 confirms the module's mute pin; the DPIA states what is captured and when	3 / 3 / 9
H-28	The instrument is read as a verdict	A failed session, or a participant inferring a diagnosis from the recording	Distress, false confirmation or false refutation of their experience	4	4	16	No individual result is ever returned; the device performs no on-board analysis; the participant card and consent form state in plain words that the instrument cannot detect, confirm or refute anything about their experience; a staffed support pathway separate from the research team	Content review of IFU-EEG-014, the consent form and every public statement, against the prohibited-words list in REG-EEG-012	4 / 2 / 8
H-29	Distress with no one present	Any of H-25 to H-28 occurs at 23:00, alone	Acute psychological harm	4	3	12	SW3 red stop button with a distinct tactile feel ends the session without touching the laptop; the session can be abandoned at any point; support number on the lid card; withdrawal at any time with no reason given	Participant card content; SW3 verified at T17	4 / 2 / 8

ID	Hazard	Sequence of events	Harm	S	P	R	Control	Verification	Residual S/P/R
H-30	Use error in placement	Ear clips swapped, EMG DIN leads mis-plugged, helmet on back to front	Bad data; a wasted session; participant self-blame	2	4	8	Electrodes permanently mounted, so there are no placements to get wrong; DIN sockets colour-coded and labelled; the runner holds the session until every site is green	T11, T18; placement guide in the kit. To be re-read when the helmet's two new couplers are approved. WH-EEG-008 §3.1.2 and §3.1.3, both specified on 2026-09-02 and neither signed, put a free-hanging touch-proof socket on each ear-reference tail and a third at the halo front for the Fpz bias lead, and add a fourth disposable pad to the three the participant already places. After that change the kit carries six sockets and six plugs of one 1.5 mm DIN 42802 family, so "electrodes permanently mounted, so there are no placements to get wrong" is true of the eight scalp sites and no longer of the whole instrument, and the cross-mate of WH-EEG-008 open item 24 is a use error this row has not scored. The scoring waits on the safety reviewer and the programme lead, who between them decide whether the ear clips travel mated and captive; it is not re-scored here on an unsigned proposal	2 / 3 / 6
H-31	Battery hatch opened	Participant opens the HM-08 hatch to "check" the device	Cell damage, short, exposure to the pack	3	2	6	Quarter-turn hatch, no tool supplied; HM-09 service key deliberately absent from the participant kit; instruction not to open	Kit packing list check at T18 that no service key is present	3 / 1 / 3
H-32	Wrong headphones fitted	Participant substitutes their own	Calibrated level invalid; possible overexposure at a different sensitivity	3	3	9	Same headphone model in every kit (A-04, 32 to 64 Ω; the shipped ATH-M20x is 47 Ω) so the calibrated level is transferable; the E-29 clamp is set on the shipped model; instruction not to substitute; level logged in session metadata	Content review; the commanded level is logged in the session metadata, so a mismatch is detectable after the fact. There is no run-time readback: the F-09 STATUS frame carries no level field, so nothing tells the runner during a session	3 / 3 / 9

Rows H-03, H-06, H-08, H-10, H-13, H-19 and H-21 are **not closed**. H-03 and H-06 carry SR-01, H-08 carries SR-02, H-10 carries SR-03, H-13 carries SR-04, H-19 carries SR-05 and H-21 carries SR-07. They are the reason section 7 exists.

4. Patient auxiliary current budget

This section is the calculation RFQ-EEG-001 Rev E S-02 points to, and it is the single home of the series-resistor noise-and-flatness arithmetic that the rest of the package cites rather than recomputes. **The resistor is 68 kΩ from 2026-09-02** (ECO-EEG-024); where a 47 kΩ figure appears below it is labelled as the superseded one and kept only so the correction is legible. It is arithmetic on datasheet figures, not a measurement. Nothing in this package has been measured.

4.1 Limits and applied parts

S-02: $\leq 10 \mu\text{A DC}$ and $\leq 100 \mu\text{A AC}$ in normal condition, including the lead-off excitation; $\leq 50 \mu\text{A DC}$ in single fault. S-02 states no AC single-fault limit; IEC 60601-1 allows 500 $\mu\text{A AC}$ in single fault for a type BF applied part and this pack uses that figure, flagged as an addition the reviewer must ratify.

There are **fourteen** patient connections: eight scalp electrodes (E_Fz, E_Cz, E_Pz, E_C3, E_C4, E_T7, E_T8, E_F7), two ear references (REF_L, REF_R) and the bias lead (BIAS_EL) at J14.1–J14.11, plus three EMG leads at the DIN 42802 sockets J15, J16, J17. That is $8 + 2 + 1 + 3 = 14$, and fourteen is the number every document in the package uses; where another document says thirteen it is wrong and is corrected. The two spare EOG channels (EOGIN1/2 at J22) are protected identically but are not fitted to a panel socket in a standard build; they are excluded from the sum and included in the fault analysis.

Every one of the fourteen passes through the same network: a **68 kΩ** 0.1 % 25 ppm thin-film resistor (R1–R16, Vishay TNPW060368K0BEEA – **corrected 2026-09-02**, ECO-EEG-024; it was 47 kΩ TNPW060347K0BEEA until that date), a BAV99 dual clamp to AVDD/AVSS (D1–D16, Nexperia BAV99,215 – BAT54S is not approved in this position) and a 10 nF C0G capacitor to AGND_REF (C1–C16, Murata GCM1885C1H103JA16D). The analogue rails are **AVDD = +2.5 V** and **AVSS = –2.5 V** referred to AGND_REF, which is the analogue mid-rail and not ground.

As of 2 September 2026 that is true of the topology as well as of the values. Until that date channel 11 was wired as an exception: D11.3 and C11.1 sat on BIAS_EL, the patient side of R11, so the clamp and the filter of the bias channel were in front of its protection resistor rather than behind it. The override in `tools/design.py` that did it is deleted and the row is generated by the generic loop, so channel 11 now reads R11.1 = BIAS_EL (patient), R11.2 = BIASOUT (module), D11.3 = BIASOUT, C11.1 = BIASOUT and C11.2 = AGND_REF, which is the wiring of the other fifteen. R11 is 68 kΩ with the rest of them. **C11 stays fitted**, at the same 10 nF C0G Murata GCM1885C1H103JA16D: RFQ-EEG-001 Rev E E-07 requires sixteen networks with C1–C16 all 10 nF C0G, and depopulating one of them would breach a mandatory line of the top-level specification silently. What the change does to the fault analysis is SF-1a and SF-6a of section 4.4; the finding it closes in the design is **SR-12**, whose disposition is still the reviewer's to sign.

J15 to J17 are named in `design.py` as a Stäubli SLB1,5-F class part rather than a confirmed PCB-mount catalogue item, and no touch-proof 1.5 mm socket with a PCB signal pin and two retention posts has yet been first-articled. That is an open sourcing item on a Class A patient-safety part and is carried in AVL-EEG-017; it does not change the arithmetic below, because the socket is a connector and not an impedance.

4.2 Normal condition, DC

Term	Source of the figure	Per connection at 25 °C	Per connection at 60 °C
BAV99 reverse leakage, one half	Nexperia BAV99,215 datasheet: $I_R \leq 100 \text{ nA}$ at $V_R = 75 \text{ V}$, $T_{\text{amb}} = 25 \text{ °C}$; $I_R \leq 2.5 \text{ }\mu\text{A}$ at $V_R = 75 \text{ V}$, $T_j = 150 \text{ °C}$	100 nA	246 nA , interpolated
ADS1299 input bias	Datasheet 300 pA typical; no maximum is specified, so the budget uses a 1 nA engineering bound	1 nA	1 nA
C0G filter capacitor insulation leakage	$IR \geq 10 \text{ G}\Omega$ at 2.5 V	0.25 nA	0.25 nA
OPA4376 inputs	0.2 pA typical, and not patient-connected: U1–U3 sit in the envelope path only	0	0
Per connection		0.101 μA	0.251 μA

The 60 °C figure is interpolated between the two datasheet points on the assumption that leakage doubles every 26.9 K, which is what those two points imply: $25 \times$ over 125 K. No credit is taken for the fact that the clamp sees $\pm 2.5 \text{ V}$ rather than the 75 V of the datasheet condition, which would reduce the figure by roughly the square root of the voltage ratio. The budget is therefore conservative by a factor of about five on its dominant term. It also assumes one diode of each BAV99 leaks at maximum while its partner leaks nothing, so no cancellation is claimed.

Aggregated as the standard measures it – the measuring device between one patient connection and the other thirteen tied together, every one of the thirteen sourcing in the same direction:

25 °C: $0.101 + 13 \times 0.101 = 1.41 \text{ }\mu\text{A}$

60 °C: $0.251 + 13 \times 0.251 = 3.51 \text{ }\mu\text{A}$ against 10 μA

Margin: 2.85x at 60 °C. The 60 °C internal temperature is an assumption for a pod dissipating 0.6 W; it must be confirmed by the 3 h thermal type test, and the reviewer should treat the DC budget as provisional until it is.

4.3 Normal condition, AC

Term	Working	Value
Lead-off excitation (E-06)	6 nA AC at 7.8 or 31.2 Hz, enabled on channels 1–8 of device 1 only (LOFF_SENSP 0xFF); worst case all eight sum through one connection	0.048 μA

Term	Working	Value
Displacement current across the isolation barrier	ADuM4160 C_I/O ≈ 2.2 pF (datasheet) plus an estimated 15 pF of module PCB and connector stray, budgeted at 20 pF. Worst case a mains-referenced host at 230 V RMS, 50 Hz: $I = 2\pi \cdot 50 \cdot 20e-12 \cdot 230$	1.45 μ A
Total		1.50 μA against 100 μ A

Margin: 67x. The 15 pF stray is an estimate, not a measurement: the module is a generic breakout with an uncontrolled layout. Even at ten times that figure the total is 15 μ A and the budget still closes.

4.4 Single-fault conditions

Two bounds are given for each fault. **Bound A** is the bare-resistor bound: the fault voltage divided by one series resistor, with a zero-impedance return, and no measuring device. It is not what IEC 60601-1 measures, and it is the number this pack judges against, because the reviewer should be shown the worse case. **Bound B** is the current the standard's own measurement would read: through the series resistor, through the figure-12a measuring device (10 k Ω at DC), and back through the other thirteen resistors in parallel.

Restated 2026-09-02 for ECO-EEG-024. At the 68 k Ω now fitted, bound B's loop is $68 + 10 + 68/13 = 83.231$ k Ω . Until that date the resistor was 47 k Ω and the loop was $47 + 10 + 3.615 = 60.615$ k Ω ; both of those figures are superseded as statements of the design and are quoted below only in the "was" column.

#	Fault	Working	Bound A	Bound B	Verdict
SF-1	One BAV99 half fails short, AVDD on the module-side node	2.5 V / 68 kΩ; 2.5 V / 83.231 kΩ	36.8 μA	30.0 μA	Passes both bounds: 26 % margin on bound A, 1.67× on bound B. Restated 2026-09-02 (ECO-EEG-024). Until that date the working was 2.5 V / 47 kΩ and 2.5 V / 60.615 kΩ, the bounds were 53.2 μA and 41.2 μA , and this row read “fails bound A by 6.4 %”. Those two figures are superseded. Nothing has been measured, and SR-01 is open for the reviewer
SF-2	One BAV99 half fails short, AVSS on the node	mirror of SF-1	−36.8 μA	−30.0 μA	as SF-1. Was −53.2 μA / −41.2 μA at 47 kΩ; superseded 2026-09-02
SF-3	One electrode conductor shorts to the WH-01 drain wire	same source, same 68 kΩ (WH-EEG-008 §5.5)	36.8 μA	30.0 μA	as SF-1. Was 53 μA / 41 μA at 47 kΩ, which WH-EEG-008 §5.5 called “over the limit”; superseded 2026-09-02, and this fault now sits inside the 50 μA limit like every other
SF-4	R _n fails short, clamp intact	clamp leakage alone, unlimited by R	0.251 μA	0.251 μA	Pass, 200× margin. Breaks the RF filter and the noise budget; a functional failure, not a safety one
SF-5	R _n fails open	no path	0	0	Pass. Channel reads open at lead-off; the contact light stays red

#	Fault	Working	Bound A	Bound B	Verdict
SF-6	C_n fails short to AGND_REF	no source	0	0	Pass. Dead channel, caught at T8 and T10
SF-6a	C11 fails short to AGND_REF – closed by the design change of 2026-09-02	C11 now sits on BIASOUT with the other fifteen filters, so this short ties the module's bias output to the analogue mid-rail and reaches the participant's Fpz electrode only through the full 68 kΩ of R11 (47 kΩ until ECO-EEG-024 was applied on 2026-09-02)	0 (no source)	0	Pass, exactly as SF-6. Nothing is injected, the bias drive is dead, and the channel is caught at T8 and T10: a functional failure, not a safety one. Superseded on 2026-09-02. Until that date this row read that C11 sat on BIAS_EL, the patient side of R11, so that the short was “a galvanic bond from the Fpz bias electrode to the analogue mid-rail with NO series resistance”, that it was “not a dead channel”, and that after it the IFU-EEG-014 §14 sentence “There is no earth connection anywhere in this instrument” was false. It was referred to the safety reviewer as SR-12. The move closes it on its own: with C11 on BIASOUT no fault of this capacitor bonds an electrode to the mid-rail except through the full series resistor, and the IFU sentence survives the fault

#	Fault	Working	Bound A	Bound B	Verdict
SF-1a	One D11 BAV99 half fails short – closed by the design change of 2026-09-02	D11 now sits on BIASOUT, so a shorted half puts AVDD or AVSS on the module-side node of channel 11 and reaches the Fpz electrode through R11, exactly as on the other fifteen channels. ECO-EEG-024 was applied the same day , so bound A is 2.5 V / 68 kΩ and bound B is 2.5 V / 83.231 kΩ; at the 47 kΩ this row was first written against they were 53.2 μA and 41.2 μA	36.8 μA	30.0 μA	This fault is now the ordinary SF-1 and is judged with it: it passes both bounds like every other channel, so it carries no figure and no finding of its own and is covered by SR-01. Superseded on 2026-09-02. Until that date D11 sat on BIAS_EL, the patient side of R11, so this row read “AVDD or AVSS reaches the Fpz electrode with NO series resistor in the path”, bound A was unbounded for want of a series element, bound B was 2.5 V / 13.615 kΩ = 183.6 μA , and the row failed S-02 on both bounds, bound B by 3.7×, on the one driven conductor that terminates on a forehead. That 183.6 μA figure is withdrawn , here and wherever else in the package it is quoted as the consequence of energising BIAS_EL – including as the stated consequence of a mis-mate into a bias socket. The bound on BIAS_EL is now the ordinary SF-1’s

#	Fault	Working	Bound A	Bound B	Verdict
SF-7	AVDD rail rises to 5 V through a module regulator fault	the clamp is reverse-biased at rest and does not conduct; the electrode node stays at AGND_REF	0	0	Pass. A rail fault alone injects nothing

#	Fault	Working	Bound A	Bound B	Verdict
SF-8	V5V or DVDD3V3 bridged onto an electrode node	5.0 V / 68 k Ω = 73.5 μ A if it could happen – 106 μ A at the 47 k Ω fitted until 2026-09-02; either way over the limit, which is why the row rests on the exclusion and not on the resistor	–	–	Excluded as not credible. The zone split at x = 62 mm, the ELECTRODE netclass (0.30 mm track, 0.40 mm clearance) and the 0.35 mm minimum electrode clearance mean no electrode net is ever adjacent to a rail above ± 2.5 V on any of the four layers. The two inner planes left of x = 62 mm are AGND_REF, not a supply, so the copper directly above and below every electrode trace is the analogue mid-rail. This is why the bipolar rail choice is a safety control and not only a signal-chain one. Verified in kicad/EEG-CAR-01_RevB_DRC_report.txt, which records zero digital nets in the analogue zone and no clearance violation anywhere on the board, the 0.35 mm electrode rule included. That is an automatic check on an unreviewed routing (SR-11), and it is re-verified on the routed Gerbers of whichever revision is finally released
SF-9	An LED conductor contacts an	3.3 V through R70–R77 (1 k Ω) = 3.3 mA	–	–	Excluded as not credible after ECO-EEG-014: two separate cables, two

#	Fault	Working	Bound A	Bound B	Verdict
	electrode conductor				connectors, two glands. No series resistance in the light drive can control this fault – raising R70–R77 to 47 kΩ would give 70 μA and still fail, and 100 kΩ would extinguish the lights. (Those are hypothetical values for the LED resistors and are not R1–R16, which ECO-EEG-024 took to 68 kΩ; R1–R16 are not in this fault path at all, because the contact is inside the helmet on the patient side of them.) The control is physical separation and its verification, nothing else. Strengthened at the site on 2026-09-02, and not eliminated. The exclusion rested on ECO-EEG-014's two cables, two connectors and two glands, and ASM-EEG-007 open item 20 recorded that the frame as built did not support it at the electrode: hm04 () cut the contact-light window as ONE box straight through a 12.40 mm body, so its inboard and outboard openings were the two ends of a single undivided cavity, and any scheme putting the electrode conductor in one end and the

#	Fault	Working	Bound A	Bound B	Verdict
					switched LED in the other had them sharing open, gel-flushed volume at all eight sites. tools/mech_gen.py hm04() now cuts two pockets, 3.20 mm wide, with 1.60 mm of printed PA12 between them – outboard the LED seat, inboard the conductor run – and the released mesh is regenerated (HM-04 model volume 1.86 cm³, mech/MANIFEST.json). The separation is now made of material rather than of distance, which is the only form of it this fault respects. What is still owed before the row may be called verified: nothing has been printed; the LED seat, its two lead passages, the dressed conductor exit and the HM-04A anchorage are still proposals in WH-EEG-008 §3.1.1 and not released geometry; both pockets open onto the outside faces of the body and the gel path runs through the same region; and no insulation or creepage figure across the 1.60 mm web has been calculated or

#	Fault	Working	Bound A	Bound B	Verdict
					measured. The control is unchanged in kind – physical separation and its verification – and only the separation half has moved
SF-10	Isolation barrier fails short, host on mains	The host's own touch current flows through the participant. A Class II supply is limited to 250 μ A to earth, implying ≈ 3.46 nF of Y-capacitance and a 919 k Ω source; the 68 k Ω series resistor adds 7.4 % to that source, giving 230 V / 987 k Ω . <i>Restated 2026-09-02: at the 47 kΩ previously fitted this read "47 kΩ adds 5 %" and ≈ 237 μA</i>	≈ 233 μ A AC	–	Within the IEC 60601-1 500 μ A AC single-fault allowance for a type BF applied part. S-02 states no AC single-fault limit and must be amended – SR-06
SF-11	Both charge interlocks fail, participant charges while wearing	As SF-10, with the consumer adaptor's secondary in place of the host	≤ 250 μ A AC	–	As SF-10. The interlocks reduce probability, not severity
SF-12	Headphone amplifier fault puts DC on the 3.5 mm sleeve	HP_GND is a carrier net referenced to DGND, inside the floating island; the ear cushions are insulating	0	0	Pass. No mains reference exists on that conductor

#	Fault	Working	Bound A	Bound B	Verdict
SF-13	An electrode net shorts to an inner reference plane through a lamination or via defect	The plane under every electrode trace is AGND_REF, the analogue mid-rail, so the fault source and the electrode reference are the same node	0	0	Pass on safety, and this is a consequence of the four-layer change worth stating: the new copper the change added under the electrodes is at 0 V analogue, not at a supply. It is a functional failure – a dead or shorted pair of channels, caught at T8 and T10

What the channel 11 change closes, and what it leaves. It closes SF-1a and SF-6a, and it closes them by moving two components rather than by argument: the fault paths those two rows described do not exist on the board that `tools/design.py` now generates. It closes SR-12 **in the design only**. The finding belongs to the safety reviewer of section 7, section 11 records its disposition as theirs to write, and this pack may not record it as closed on the strength of the programme having taken the option the analysis recommended. Two things travel with the change and neither is a patient-current fault.

The one thing the package cannot compute. With C11 on BIASOUT the series resistor no longer isolates the module's bias amplifier from 10 nF of C0G at its own pin, and no document here establishes that the amplifier is stable driving it. The module is not bought, and the bias loop closes inside it: ICD-EEG-006 §1.1 records at J2 that BIASOUT is an output of the module and that BIASIN is driven by nothing on the carrier. No ADS1299 datasheet figure answers for a third-party module's output. It is therefore declared where this package declares everything it assumes about that module: as a row in **ICD-EEG-006 §2.1's assumption table**, "Bias amplifier stable driving 10 nF C0G at BIASOUT, with 68 kΩ in series to the electrode", with the consequence if untrue that the bias loop peaks or oscillates and common-mode rejection degrades on all sixteen channels. That is an incoming-qualification condition on a supplier not yet chosen and a first-article scope check, and it is the **module-qualification owner's** to accept before any PIEEG-8-class module is qualified, and to carry into the RFQ pack. The same §1.1 note is itself owed a correction: it still records D11 and C11 as sitting on BIAS_EL, "the patient side of it, where the other fifteen have their clamp and filter on the module side", and cites SF-1a, SF-6a and SR-12 for the consequence. That sentence is superseded on 2026-09-02 and its repair belongs to ICD-EEG-006's owner, not to this pack. The safety exposure if the assumption is wrong is bounded and functional, not a fault current: an oscillating bias amplifier is rail-limited and reaches the forehead only through the 68 kΩ, so at most about 37 μA AC – it was about 53 μA at 47 kΩ – inside the 100 μA AC normal-condition allowance of section 4.3 for a type BF applied part, and it presents as degraded CMRR on a bench rather than as a hazard on a head. If first article shows peaking, the cheap directions are all still open with the pad populated: depopulate C11, or drop to the 4.7 nF C0G same-series alternate that AVL-EEG-017 already approves on

a written ECO, or fit ≤ 1 nF behind a 10 k Ω isolation resistor as a schematic change. The expensive direction is the other one, which is why C11 is fitted now.

The 183.6 μ A is withdrawn, and nothing else closes with it. Any document that cites 183.6 μ A as the consequence of putting an electrode or a fault on BIAS_EL is superseded from 2026-09-02, including the mis-mate case raised against a free-hanging bias socket – a K2 ear clip or a K3 EMG lead entering it now lands on a node bounded by the ordinary SF-1, not on an unresisted one. The mis-mate itself is a use-error item and stays open where it is raised, in WH-EEG-008's open item 24 and in the register of connector cross-mates; it is not closed here and it is not closed by this change. Nor did touch-proofing that interface ever close SR-12: a connector does not alter what flows out through an electrode. The topology change is what closes SR-12 in the design, and the reviewer's signature is what closes it in this pack.

4.5 Conclusion and the margin

Limit	Calculated	Margin	Closes?
Normal, DC ≤ 10 μ A	3.51 μ A at 60 °C	2.85×	Yes
Normal, AC ≤ 100 μ A	1.50 μ A	67×	Yes
Single fault, DC ≤ 50 μ A	36.8 μA (bound A) / 30.0 μ A (bound B)	1.36× / 1.67×	Yes, on both bounds, in the design. <i>Restated 2026-09-02: at the 47 kΩ fitted until that date this row read 53.2 μA / 41.2 μA, –6.4 % / 1.21×, and “No, on bound A”. It is met by calculation, it is not measured, and SR-01 is open for the reviewer</i>
Single fault, AC (not stated in S-02)	≈ 233 μ A	–	Limit missing

SR-01 – the finding that matters. Restated 2026-09-02, and still open.

What it said until that date. With ± 2.5 V rails and 47 k Ω , the single-fault DC budget did not close on the bare-resistor bound: $2.5 \text{ V} / 47 \text{ k}\Omega = 53.2 \mu\text{A}$ against the 50 μ A limit of S-02, a 6.4 % overshoot. It closed on the measurement the standard actually prescribes, but only by 17 %. That overshoot stood through three revisions.

What is true now. **ECO-EEG-024 is applied.** R1–R16 are 68 k Ω , so $2.5 \text{ V} / 68 \text{ k}\Omega = 36.8 \mu\text{A}$ on bound A and $2.5 \text{ V} / 83.231 \text{ k}\Omega = 30.0 \mu\text{A}$ on bound B, against 50 μ A. **S-02 is met in the design, on both bounds**, and this pack says so plainly because it is what the arithmetic gives. The disposition the pack recommended below – option 2 – is the one the programme took.

Why the finding is still open. Three things did not change when the resistors did. **Nothing has been measured:** the figure is arithmetic on datasheet values, the interpolated BAV99 leakage among them, with no measurement anywhere in the

package and no accredited laboratory available to the programme. **No unit exists** to measure. And **the finding belongs to the safety reviewer**, not to the programme: a design change made by the people who wrote the analysis is not a disposition, and section 7 exists precisely so that someone outside the programme signs one. SR-01 therefore stays **Open** in section 10, and section 7.3 item 4.3 is the form it closes on. What the reviewer is asked has changed, and that is the whole of the change: they are handed a design that **meets** the limit and asked to confirm it, rather than one that does not and asked to accept it. The three dispositions are kept below because the reviewer may still direct any of them:

1. **Accept bound B.** Record that the compliant measurement reads 41.2 μA calculated, accept the 17 % margin, and require T23 on every unit as the routine evidence. No change to the board. Cheapest and weakest.
2. **Raise the series resistance to 68 k Ω . THIS IS THE DISPOSITION THE PROGRAMME TOOK, on 2026-09-02, as ECO-EEG-024; it is applied in tools/design.py and awaits the reviewer's ratification, not their selection.** R1–R16 are **68 k Ω 0.1 % thin film** in the same 0603 footprint with the already-approved **10 nF C0G** capacitor, so sixteen parts changed and nothing moved. $2.5 \text{ V} / 68 \text{ k}\Omega = \mathbf{36.8 \mu\text{A}}$, a 26 % margin under the 50 μA limit. The consequences, all calculated:

Property	47 k Ω / 10 nF (superseded 2026-09-02)	68 k Ω / 10 nF (ECO-EEG-024, fitted)	Requirement
Single-fault bound A	53.2 μA	36.8 μA	$\text{S-02} \leq 50 \mu\text{A}$
Single-fault bound B	41.2 μA	30.0 μA	$\text{S-02} \leq 50 \mu\text{A}$
Corner frequency	339 Hz	234 Hz	–
Loss at 100 Hz	–0.36 dB	–0.75 dB	E-10 states both branches: ± 0.5 dB at 47 kΩ and ± 1.0 dB at 68 kΩ. ECO-EEG-024 is taken, so the ± 1.0 dB branch is the one in force. Both states are stated together wherever E-10 appears
Johnson noise, 0.5–70 Hz	0.233 μV	0.28 μV	–
Total input-referred noise with the ADS1299's 0.14 μV	0.27 μV	0.31 μV	$\text{E-03} \leq 1.0 \mu\text{V}$

Property	47 k Ω / 10 nF (superseded 2026-09-02)	68 k Ω / 10 nF (ECO-EEG-024, fitted)	Requirement
DC offset from 250 nA clamp leakage	11.8 mV, 6.3 % of range	17.0 mV, 9.1 % of the ± 187.5 mV range at gain 24	headroom only

The noise cost is 0.04 μ V and the budget still has more than 3 $\times$ of margin. The real cost is passband flatness: 68 k Ω into the same 10 nF pulls the corner down to 234 Hz and costs 0.75 dB at 100 Hz. **E-10 already states both branches and permits ± 1.0 dB for exactly this case**, so taking the disposition moved the requirement onto that branch rather than breaching it; ECO-EEG-024 carries the widening with the resistor change, and TST-EEG-004 T22 grades against ± 1.0 dB on the 68 k Ω build. The wider band is earned by the 68 k Ω and by nothing else. Rev A of this pack proposed 100 k Ω with the 4.7 nF alternate instead; that reaches 25 μ A but changes thirty-two parts, changes the approved capacitor value and raises the noise to 0.37 μ V, and 68 k Ω buys enough margin for less disturbance. **The ruling is 68 k Ω , recorded as ECO-EEG-024.**

3. **Requalify the clamp.** Return to a low-leakage TVS (Nexperia PESD5V0X1BL class, the part the Rev A internal BOM carried) and keep 47 k Ω . This does not help: the fault is a *short*, not leakage, and 2.5 V across 47 k Ω is 53.2 μ A whatever the clamp is. Recorded here so the reviewer does not have to rediscover it.

What is actually built. Restated 2026-09-02. *Until that date this paragraph read that the two Phase 1 prototypes were “fabricated and assembled with 47 k Ω fitted” while ECO-EEG-024 stayed open. Both halves of that are now wrong: **no prototype has been fabricated or assembled** – nothing in this package has been built – and **ECO-EEG-024 is applied, not open**.* What is true is that tools/design.py fits 68 k Ω , that the carrier BOM and the fabrication data carry 68 k Ω , that ECO-EEG-024 is recorded in the register ECO-EEG-016 against EEG-CAR-01 Rev B, RFQ-EEG-001 E-07 and E-10, the carrier sub-BOM and TST-EEG-004 T22’s calculated expectation, and that E-10 therefore stands on its ± 1.0 dB branch. The two Phase 1 prototypes, when they are built, are built with 68 k Ω . **The measurement of the real single-fault current on a Phase 1 unit is still made before the Phase 2 build is released, and the reviewer’s disposition is taken then.** The design meets S-02; the *finding* SR-01 stays open until the reviewer signs and the measurement exists.

4.6 How the limit is verified and who signs it

Routine per-unit evidence is TST-EEG-004 **T23**, fourteen DC and fourteen AC readings, host connected and disconnected, plus fourteen single-fault DC readings – seventy values in the unit record. T23 states its own limitation plainly: it uses a 100 k Ω measuring resistor rather than the figure-12a network and applies one single-fault condition. It is a stand-in.

The real verification of S-02 is a type test with an MD-2 measuring device per IEC 60601-1 figure 12a applied at each of the fourteen terminations, normal and single fault, on one Phase 1 unit, witnessed by the safety reviewer. It does not exist yet and the pack does not pretend otherwise. **Sign-off: the safety reviewer, on the section 7 form, item 4.3.**

5. Creepage, clearance, and why the carrier has none to check

5.1 Where the barrier is

The barrier is entirely inside the ADuM4160 module at J10. The host receptacle is on the module – USB-B on the only named candidate, brought to a panel USB-C by the WH-09 pigtail, which is a live non-conformance against E-24 and is stated as such in section 5.4. The four carrier pins are VDD_IS0, USB_DN, USB_DP, DGND – every one of them on the device side. There is no host-side net anywhere in design.py's netlist, on any of the four layers, at any coordinate. **The carrier has no isolation barrier, and therefore has no creepage or clearance across one to assess.** That is the whole isolation strategy in one sentence, and it is the reason the carrier needs no milled creepage slot: not because of how many layers it has, but because no conductor on it is ever mains-referenced.

The layer count changed during layout and the reviewer should know what it does and does not change here. EEG-CAR-01 Rev B is a **four-layer** board – L1 signal, L2 reference plane, L3 reference plane, L4 signal, with AGND_REF left of x = 62 mm and DGND right of it on both inner layers, stitched together only at the R90 star point. Package v1 argued that two layers would be enough; doing the layout showed that on two layers the bottom side would have to be both the reference plane and the second routing surface, and it cannot be both. For the isolation case the four-layer change is neutral in principle and adds work in practice: there are now **four** copper layers on which the keep-out under the barrier must be proved, and two of them cannot be inspected on a finished board. Two properties make that verifiable rather than hopeful. Vias are **through vias only** – no blind, buried or back-drilled structures – so every hole in the barrier region appears in the drill file and on both outer plots. And the keep-out is encoded as a rule area on all four copper layers, so the DRC proves it rather than a person measuring a plot. In the Rev B routing it does: kicad/EEG-CAR-01_RevB_DRC_report.txt lists no isolation keep-out violation on any of the four layers. Section 5.5 gives that result with its figures, and states what the same report does and does not settle.

The consequence, which the reviewer should hold on to: the entire safety case for S-03 rests on a purchased breakout board whose layout the programme does not control.

5.2 What the barrier must achieve

Working voltage on the host side is mains, taken as 250 V RMS, pollution degree 2, material group IIIb unless the module vendor declares better.

Means of protection	Creepage	Air clearance	Dielectric strength	Source
1 MOOP	2.5 mm	2.0 mm	1500 V AC	IEC 60601-1 table 13/16
1 MOPP	4.0 mm	2.5 mm	1500 V AC	IEC 60601-1 table 12/11
2 MOPP	8.0 mm	5.0 mm	4000 V AC	IEC 60601-1 table 12/11

Means of protection	Creepage	Air clearance	Dielectric strength	Source
ADuM4160 in RW-16	≥ 8.0 mm external	≥ 8.0 mm external	5000 V RMS, 1 min	Analog Devices datasheet; reinforced per DIN V VDE V 0884-10, CTI ≥ 600 V (material group I)

The part itself therefore exceeds S-03's ≥ 2.5 kV RMS requirement and meets 2 MOPP distances at 250 V. What the part achieves and what the *module* achieves are different questions, and only the second one matters. The module's own barrier pad-to-pad spacing, its silkscreen and mask over the barrier, and any Y-capacitor fitted across it are all outside the programme's control.

5.3 The keep-out under the barrier

The one thing the carrier can do about a barrier it does not carry is refuse to put copper underneath the half of the module that is mains-referenced. It does. The keep-out's coordinates and layer scope are stated once, in **DSN-EEG-003 Rev D section 3.3 rule 4**, and are not restated here. What this pack adds is the safety consequence and the geometry that carries it:

Property	Value
Region	The rule area of DSN-EEG-003 §3.3 rule 4, a 9.0 × 20.0 mm rectangle reaching the x = 150.0 mm board edge
Layers	No copper on any of the four layers : no pour on L1 or L4, no plane copper on L2 or L3 , no via, no track, no test point, no silkscreen conductor. The inner planes are the part a visual check cannot see and the part the DRC rule area exists for
J10 position	(136.0, 6.0), a 1×4 socket running in +Y; pins span y = 6.00 to 13.62 mm at x = 136.0, clear of the keep-out edge
Nearest other part	J11 (ATECC608B) at (136.0, 20.0) – device side, outside the barrier region
Encoding	A KiCad rule area on all four copper layers plus a custom DRC rule, so that this one dimension is proved by the check and not by an eye. It is proved in the Rev B report, which lists no isolation keep-out violation and no violation of any other rule either. Section 5.5 says what that settles and what it does not

MP-01, the printed module plate, positions the ADuM4160 module above the carrier so that the device half sits over J10 and **the whole of the host half – the module's own host receptacle (USB-B on the named candidate, panel USB-C via WH-09, section 5.4), the host-side VBUS and ground copper, and the host-side half of the**

ADuM4160 body – lies in plan view inside the keep-out rectangle. No carrier conductor, on any layer, is beneath any of it.

5.4 The slant path, and what is still not fixed

The plan-view keep-out removes carrier copper from under the host half, but the shortest path from a carrier conductor at the keep-out edge to a host-side module conductor does not run in plan view. It runs over the edge of the keep-out and up through the standoff to MP-01. That combined slant distance must be at least the 8.0 mm of 2 MOPP at 250 V.

This is now closed. The carrier-to-plate standoff is fixed at **M3 × 18 mm nylon hex, female-female, four off at MH1–MH4, with eight M3 × 6 nylon pan screws**, and MP-01 is **146.0 × 126.0 × 3.0 mm**. The vertical component of the slant path is therefore 18.0 mm before any horizontal component is added, which is more than twice the 8.0 mm the safety case asks for, with no reliance on the horizontal offset at all. The standoff height also fits the enclosure: floor 2.5 + boss 6.0 + carrier 1.6 + standoff 18.0 + plate 3.0 + modules ≤ 18.0 = **49.1 mm** against the 55.5 mm internal depth of POD-P1, a margin of 6.4 mm. **SR-08 is closed on that ruling**, and the consequence is that the standoff height is now a risk-control dimension: section 9 lists it, and shortening it reopens the finding.

Open: the module is uncontrolled. No vendor has been fixed for the ADuM4160 breakout and no vendor board file or dimensioned drawing has been obtained. Until one is, the module's own creepage is unknown, and so is the exact position of its host-side copper relative to the keep-out. This is the weakest link in the entire safety case and is named as such. It is SR-09, and it is Critical.

Open: the isolator's host connector. The only named candidate module presents a USB-B host receptacle where E-24 asks for USB-C. The interim resolution is a short USB-B-to-USB-C panel pigtail, WH-09, until a module with a USB-C host connector is qualified. This is a live non-conformance, not a settled design, and it changes nothing about the barrier itself.

5.5 Verification

What	How	When	Who signs
Carrier has no host-side copper	Netlist inspection against design.py; copper-plot overlay of the keep-out on all four layers , inner layers included	Once per routed Gerber revision	Programme, countersigned by the safety reviewer
≥ 8 mm creepage in the artwork	Measured from the artwork with the keep-out overlay (TST-EEG-004 section 14, the creepage and clearance type test)	Once per routed Gerber revision	First-article report

What	How	When	Who signs
≥ 8 mm slant path to MP-01	The dimensioned MP-01 drawing and the 18 mm standoff of section 5.4	Once per mechanical revision	Programme, seen by the safety reviewer
Module isolation rating	Vendor certificate and part marking photograph	T00, per module lot	Incoming inspection
No path host to applied part	T20: 500 V DC, 60 s, host-side shell/VBUS/D+/D– shorted against J14.1–J14.11 and J15–J17 shorted; ≥ 1 GΩ	Every unit	Test operator
2.5 kV RMS type test	The module supplier's certificate, checked once at incoming inspection and never repeated per unit. There is no per-unit hipot on this product	T00, per module type	Safety reviewer accepts or rejects
DevKitC-1 receptacles blanked	Visual, before labelling; photograph in the unit record	Every unit	Assembly operator, first article by the reviewer

What the routing actually produced. EEG-CAR-01 Rev B is routed on four layers, 150.0 × 130.0 mm, with 3 745 track segments and 552 through vias, and each reference plane is one continuous island per net on both In1.Cu and In2.Cu. **All 145 nets are fully connected:** none is left unclosed and none is without copper. The smallest measured clearance is 0.260 mm on F.Cu, 0.275 mm on B.Cu and 0.285 mm on the planes, against a 0.20 mm rule. The narrowest conductor is 0.20 mm and the smallest plated hole is 0.30 mm. Copper stands 2.00 mm from every non-plated hole and there are no zone crossings. No digital net enters the analogue zone. There is exactly one AGND_REF-to-DGND bridge and one HARN_SHIELD-to-DGND bridge. There are no duplicate copper segments and no duplicate via positions. The report's own line is **“VIOLATIONS: 0 – none. The board passes every rule listed above.”**

The isolation rule area is clean on all four copper layers. That is said here because kicad/EEG-CAR-01_RevB_DRC_report.txt says it: the report lists no isolation keep-out violation, so no pour, plane, via, track or test point sits inside the 9.0 × 20.0 mm strip on L1, L2, L3 or L4. It is not asserted from the fact that a rule area was drawn. The reviewer is asked to read the report issued with the Gerbers that are actually released, because that report and not this sentence is the evidence, and because the two inner layers cannot be checked on a finished board.

The board passes, and passing is not the same as reviewed. The gate ECO-EEG-016 section 3 sets for releasing fabrication data is zero DRC violations, every net one connected copper island, and both inner planes continuous under the analogue zone. All three are now met, so the data in kicad/ is **released for review under RFQ-**

EEG-002A. It is **not released for fabrication** and no board may be ordered from it: the routing was produced by the programme's own tools, no human layout engineer has looked at it, and that review is what fabrication release waits on. Nothing in this package has been manufactured or measured, so every figure in this section is geometry read off the artwork, not something a board did.

The router relaxed 169 connections to get there. Thirty-six took a conductor narrower than the 0.25 mm preferred width; 133 kept full width and took a reduced gap instead. Every one is at or above the 0.20 mm minimum conductor and the 0.20 mm minimum gap, 0.35 mm on electrode nets, which is why they are relaxations and not violations, and each is named pad to pad in the DRC report. A board that closes at minimum geometry is not the same board as one that closes at preferred geometry, even when every rule passes: it leaves less room for a fabricator's own tolerances, and the thinnest places are the first ones to look at. The layout reviewer is asked to read that list and say which of the 169 must change.

That is a safety matter and not only a layout one, for three reasons. First, this is still a release gate sitting *before* the gate of section 7, though a smaller one than it was: the DRC evidence, item 7 of the receipt checklist in 7.2, is now satisfied, but no board has been fabricated, so no assembled unit exists, so item 17 cannot be satisfied and the review still cannot start. Second, the routing can still change. A layout engineer reading the 169 relaxations may move copper, and several of them are on nets this pack reasons about – AVSS, AVDD, DVDD3V3, V5V, BIASOUT and SRB1 among them – so the zoning, clearance and star-point arguments of sections 4 and 5 are confirmed today by a checker and not by a person, and must be re-checked against the routing finally released. Third, the 0.35 mm electrode clearance is a risk control, and the same report that records zero violations records it as met everywhere on the board – by the automatic check, on an unreviewed routing, with the same qualification as everything else here. This is carried as **SR-11** in section 10 and as item 4.16 of the checklist.

6. Battery safety

6.1 The cell

Property	Requirement	Reference
Type	Protected 18650, ≥ 3000 mAh, with an integral PCM giving over-charge, over-discharge and short-circuit protection	S-04, E-22, kit BOM item 11
Reference parts	Panasonic NCR18650B protected (3400 mAh, 3.6 V nominal, 12.24 Wh) or Samsung INR18650-35E in a protection holder (3500 mAh, 12.6 Wh)	kit BOM
Transport classification	Under 20 Wh per cell and under 100 Wh per battery, so UN3481, PI967 section II when installed in equipment	S-09

Property	Requirement	Reference
Documents required at incoming	UN 38.3 test summary; IEC 62133-2 certificate ; SDS; lot code	T00, AVL-EEG-017
Connector	J13, JST B2B-PH-K 2.00 mm, polarised, VBAT and DGND	design.py
Retention	The HM-10 keyed cell carrier, behind the HM-08 quarter-turn hatch; the cell cannot be inserted reversed	S-04, M-02
Thermistor	Not fitted. S-04's thermistor-monitored charging is not met – see 6.3 and SR-03	S-04

HM-07 is the boom microphone arm and is not part of the battery compartment. Any text that calls the hatch HM-07, or that merges the hatch and the carrier into one part number, is at the package v1 numbering; PARTS-EEG-019 §3.1 is the register.

No spare cell travels in the kit. DSN-EEG-002 Rev E section 7 says a charged spare travels in the case; the kit BOM Rev B carries one cell per kit and this pack rules that the BOM is correct. A loose charged spare would move the shipment from PI967 (contained in equipment) to PI966 (packed with equipment), with different packing and marking, and would put an unprotected charged cell in foam next to a metal ratchet. DSN-EEG-002 Rev E must be corrected, and it has not been: the shipped Rev E file is byte-identical to the one released in package v2.1, and ECO-EEG-016 section 1.1 records that it was not re-issued in this correction round. **Until it is, its section 7 is superseded on this point** by this section and by REG-EEG-012 Rev B section 3.1, and nobody works from it for the spare cell.

6.2 Charge control

Charging is by a **separate, charge-only USB-C receptacle** on the panel, wired through a JST PH pigtail to J24. No data conductor enters it. The path is F1 (Bourns MF-MSMF110-2 PTC, 1.1 A hold / 2.2 A trip) then D24 (Nexperia PESD5V0S2BT TVS) to VBUS_CHG at J12.3, with the R84/R85 divider producing VBUS_DET at GPIO46. R84 is 100 kΩ and **R85 is 150 kΩ**, giving $5 \times 150/250 = 3.00 \text{ V}$ at VBUS = 5 V, comfortably above the 2.48 V input-high threshold of a 3.3 V ESP32-S3 input; the 56 kΩ of the first cut gave 1.79 V and would not have asserted the first S-01 interlock reliably, which is ECO-EEG-022. This receptacle is galvanically separate from the isolated data USB-C on the ADuM4160 module: ECO-EEG-003 exists precisely because Rev A tried to make one connector do both, which cannot be reconciled with S-03.

The interlock (S-01) has two mechanisms:

1. firmware refuses CMD_START_SESSION while VBUS_DET is high, with a defined ACK code;
2. CHG_CE (GPIO47, at J12.4) holds the charger disabled for the whole of a session.

Both are verified per unit at T4 and T21. **Both are firmware.** There is no hardware gate that would keep the charger off if the firmware misbehaved, and section 3 records that as H-08 with a decision owed at SR-02. The reviewer must rule on whether a firmware-

only interlock is acceptable for a device used unsupervised, or whether an independent hardware inhibit – a two-input gate between VBUS_DET, a hardware session-active latch and CHG_CE – is required before Phase 2. It is two parts on a board that has not been fabricated.

6.3 Thermal

Case	Working	Result
Recording, Phase 1 in POD-P1	150 mA at 3.9 V (T3 limit) = 0.585 W, essentially all dissipated inside the shell. POD-P1 base 163 × 143 × 58 mm external, area 0.0821 m ² , h ≈ 8 W/m ² K natural convection plus radiation	$\Delta T \approx 0.9 \text{ K}$, calculated. 24.9 °C at 24 °C ambient; 35.9 °C at 35 °C ambient
Recording, Phase 2 occipital shell	Same 0.585 W in the smaller HM-01 occipital shell, 116 × 46 × 88 mm, area 0.0392 m ²	$\Delta T \approx 1.9 \text{ K}$, calculated. This is the bounding case and the one the margin is quoted against
Charging	Charger dissipation at 1 A from 5 V to a 4.2 V cell ≈ 0.8 W, plus 0.4 W system	$\Delta T \approx 1.8 \text{ K}$ in POD-P1, 3.8 K in the occipital shell, calculated. Never coincident with wearing (S-01)
Limit	Any surface in continuous skin contact	43 °C (IEC 60601-1 applied-part limit)

Margin at 35 °C ambient while recording, taking the worse of the two enclosures: 6 K. Verified by the 3 h thermal type test with thermocouples on the TPU brow and occiput pads, the pod surface and the cell surface. The 60 °C internal air temperature assumed in section 4.2 is intended to bound this; if the thermal run reads higher, the DC current budget must be recomputed.

E-23 forbids charging above 45 °C and S-04 asks for thermistor-monitored charging.

Neither is met, and neither can be tested on this hardware. J12 carries no TS conductor – its eight pins are VBAT, DGND, VBUS_CHG, CHG_CE, SDA, SCL, VSYS, NC_CHG_STAT – and J13 is a two-pin connector with no thermistor pin, so a cell NTC has no route to the charger. There is no NTC net anywhere in design.py. TST-EEG-004's traceability matrix records E-23 as partial for exactly this reason, and no document in this package may record S-04 or E-23 as met. This is SR-03; section 7 item 4.5 is where it is decided.

6.4 Shipping

Every despatch, outbound and return, is UN3481, PI967 section II: cell installed in equipment, state of charge ≤ 30 %, cell retained so it cannot move or short, lithium battery mark on the outer carton, correct carrier documentation, and no air transport unless the carrier accepts section II by air. **PKG-EEG-015 section 7 carries the procedure and the return-leg paperwork**, because the return shipper is an untrained participant in their own home and the programme cannot delegate that obligation to them; this pack states the obligation and does not restate the procedure. The programme claims no marking exemption.

6.5 A kit returns with a damaged cell

This will happen. The procedure:

1. **At the door.** Goods-in opens every returning case in a designated area away from combustibles, with a lithium-fire bucket to hand. Any case that is hot, deformed, smells, or was reported as dropped is not opened indoors.
 2. **Quarantine on sight.** A swollen, dented, discoloured or leaking cell, or a cell whose wrapper is torn, is a **quarantine, no-rework** disposition under QP-EEG-010. The cell is not charged, not tested and not re-used. It goes to a metal container with a lid, in sand, outside, pending disposal through the programme's WEEE and battery route.
 3. **The kit follows the cell.** A kit that carried a damaged cell does not re-enter circulation on a visual check. It gets the full incoming subset (T3, T7, T8, T10, T13, T17), plus T20 insulation resistance, plus an inspection of J13, the HM-10 carrier, the carrier board around J13 and the HM-08 hatch, before it is released.
 4. **It becomes evidence.** Every damaged-cell return is a nonconformance that feeds this risk file's post-production loop (section 7.6). Two in a fleet year is a signal about the case, the foam or the retention, not about one participant.
 5. **The participant is told they did nothing wrong**, in those words, before anything else is said to them. On this cohort, a message that reads as blame for damaging a device that "records the brain" is a harm in itself.
-

7. The safety review

This is the gate named in RFQ-EEG-001 Rev E sections 3 and 9.2 and in DSN-EEG-003 Rev D section 8 item 1. **Phase 2 is not released without a signed record from this section, and nothing goes on a head before it.** Fabrication, assembly and quoting are not blocked by it.

Something else blocks the *start* of this review, and the reviewer should know it before reading further. The Rev B routing now meets all three conditions of the ECO-EEG-016 section 3 gate – zero violations, all 145 nets one connected copper island, and both inner planes continuous under the analogue zone – so the fabrication data is **released for review under RFQ-EEG-002A**. It is still **not released for fabrication**, because no human layout engineer has read the routing and that review is the scope of RFQ-EEG-002A. No board has therefore been made and no unit assembled, so item 17 of the receipt checklist below cannot be met. That is a release gate ahead of this one. It is stated in section 5.5 with its figures and carried as SR-11 in section 10. This review does not start until a unit has been built from a released Gerber set and tested.

7.1 Who is qualified to do it

A chartered or registered electrical engineer with demonstrable medical-electrical experience – IEC 60601-1 design review or type testing – who is **independent of the design author**. Independence means: no involvement in producing DSN-EEG-003, design.py, the schematic set or the firmware; no financial interest in the manufacturer; not a member of the programme. A CV is attached to the record and a signed declaration of independence is item 1 on the form.

A second reviewer is not required. If the programme cannot engage a reviewer meeting this definition, it does not proceed to Phase 2 with a weaker one; it stops at Phase 1.

7.2 What the reviewer is given

A receipt checklist the programme must satisfy **before** the review starts. A missing item is not a note in the report, it is a reason to postpone.

#	Input	Document
1	This risk file	RISK-EEG-011 Rev B
2	Requirements	RFQ-EEG-001 Rev E, sections 4, 5, 7, 8, 9
3	Design package and the ECO history	DSN-EEG-003 Rev D, ECO-EEG-016
4	The design source of truth	tools/design.py at the reviewed commit hash
5	Schematic set, 8 sheets	SCH-EEG-005 Rev B
6	Interface control – every module pinout and jumper	ICD-EEG-006 Rev C
7	Routed Gerbers for all four layers, drill, CPL, the stack-up drawing, and the DRC report for the released revision, showing zero remaining violations and the isolation rule area clean on all four layers. Satisfied as at the Rev B report of 2026-09-02, which is the report produced after the channel 11 topology change was routed in: zero violations, 145 of 145 nets connected, 3 745 segments and 552 vias, rule area clean on all four layers. <i>Corrected from “the Rev B report of 2026-09-01”, which the re-route superseded.</i> It is given to the reviewer with its list of the 169 relaxed connections, and is replaced by the report for whatever revision the layout review of RFQ-EEG-002A finally releases if that review changes the routing (SR-11)	kicad/EEG-CAR-01_RevB_DRC_report.txt, superseded by the released revision from RFQ-EEG-002A
8	Harness and cable definition	WH-EEG-008 Rev B, especially §5.5

#	Input	Document
9	Helmet, applied parts and fitting	DSN-EEG-002 Rev E, sections 3–7. Not re-issued since package v2.1 and byte-identical , so read section 7 against two supersessions and do not work from it: it says a charged spare cell travels in the case, which section 6.1 above and REG-EEG-012 section 3.1 rule out, and it describes the battery hatch as interlocked to the session state, which REG-EEG-012 section 3.4 states does not exist in this design and may not be claimed. Its section 11 travel-case figure FIG-19 is the two-layer foam and is superseded by the CASE-00 Rev C schedule of PKG-EEG-015 section 2.2. Sections 3 to 6 are current
10	Mechanical drawings: MP-01, POD-P1, the standoff stack of 5.4	MECH set, PARTS-EEG-019
11	Production test and its traceability matrix	TST-EEG-004 Rev C
12	Quality plan and disposition rules	QP-EEG-010 Rev B
13	Regulatory file, declaration and DPIA	REG-EEG-012 Rev B
14	Module datasheets and certificates: ADuM4160 (J10) isolation certificate and vendor board file, charger module (J12), both ADS1299 modules, ATECC608B, buck-boost	AVL-EEG-017 evidence pack
15	Cell dossier: UN 38.3 summary, IEC 62133-2 certificate, SDS, PCM datasheet	AVL-EEG-017
16	Biocompatibility declarations for every material row	REG-EEG-012 §6.2
17	One fully assembled unit , with its complete TST-EEG-004 record and calibration certificate	the unit

#	Input	Document
18	Participant-facing material as it will actually be printed	IFU-EEG-014, consent form

7.3 The checklist

Each item has an objective acceptance criterion, a place for evidence, and a verdict of **Pass / Fail / Not applicable / Deferred with condition**.

#	Question	Acceptance criterion
4.1	Is the applied-part classification correct and argued?	The 14 patient connections are treated as one type BF applied part; the argument is written down and the reviewer agrees with it
4.2	Is a firmware-only charge interlock acceptable? (SR-02)	Either accepted in writing with the reasoning, or an ECO for an independent hardware inhibit is raised and Phase 2 is conditional on it
4.3	Is the section 4 auxiliary-current budget accepted, now that S-02 is met in the design at 36.8 μ A (bound A) and 30.0 μ A (bound B) against 50 μ A, on calculation alone with nothing measured? (SR-01) <i>Restated 2026-09-02: this question read "given that S-02 is not met at 53.2 μA against 50 μA on bound A" while 47 kΩ was fitted.</i>	A written disposition of SR-01. ECO-EEG-024's 68 k Ω has been applied and E-10 sits on its ± 1.0 dB branch, so the reviewer ratifies or directs otherwise rather than choosing from three open options – the other two of section 4.5 remain available to them. The values of R1–R16, C1–C16 and D1–D16 released for Phase 2 are ratified; E-07's "value confirmed at safety review" is closed; the Phase 1 measurement is scheduled, because no figure in section 4 has been measured
4.4	Is the isolation case accepted? (SR-06, SR-09)	Module certificate seen; keep-out verified on the routed copper plot of all four layers ; the 18 mm standoff seen on the dimensioned MP-01 drawing; T20 witnessed on the assembled unit; a decision recorded on whether one reinforced barrier plus 68 k Ω limiting is sufficient here, and on proceeding with an unfixed module vendor
4.5	Battery, charge and temperature (SR-03)	Cell dossier complete; HM-10 carrier and HM-08 hatch inspected on the unit; a disposition on the missing thermistor path, which today leaves S-04 and E-23 not met – board change, module-side NTC, or an accepted reduced control with a stated instruction

#	Question	Acceptance criterion
4.6	Biocompatibility (SR-05)	A declaration or a documented history-of-safe-use argument for every material row in REG-EEG-012 §6.2, valid after 25 reprocessing cycles
4.7	EMC and radio	CISPR 11 class B pre-scan report seen; T24 result and firmware hash on the reviewed unit seen
4.8	Harness safety	Two-cable separation confirmed by inspection of the built helmet; insulation resistance record seen; SF-3 of section 4.4 ruled on together with SR-01
4.9	Acoustic ceiling (SR-04)	E-29's 100 dB SPL ceiling exists as a numbered requirement; the reviewer sees the artificial-ear type test result of TST-EEG-004 T28 (TST section 14), or accepts in writing that the firmware clamp is the only control against a calculated 110 dB SPL full-scale output, with the residual stated
4.10	Mechanical and thermal	M-04 drop report seen with the post-drop re-test; 3 h thermal report seen; no skin-contact surface above 43 °C
4.11	Labelling and instructions	M-03 label on the unit matches the calibration record and the USB iSerial in the TI0V-B-nnnn form of PKG-EEG-015 section 5; the participant card carries the stop instruction, the do-nots, the support number and the not-a-medical-device sentence verbatim
4.12	Residual risk acceptability	Every row of section 3 with $R \geq 10$ either reduced or accepted in writing; the section 8 list matches what the participant is actually told
4.13	Risk-control components	The section 9 list is complete, and QP-EEG-010's no-substitution rule covers every entry, the 18 mm standoff included
4.14	Reprocessing (SR-07)	The 25-cycle validation is either complete or scheduled with Phase 2 release conditional on it

#	Question	Acceptance criterion
4.15	The ENV_CMP re-reference (SR-10)	ECO-EEG-023 moves U7's supply to DVDD3V3/DGND and AC-couples its inputs onto a DVDD3V3/2 divider, so an analogue-referenced signal now drives a digital-referenced comparator. The reviewer confirms that no new path exists between the analogue island and DGND other than R90, and the layout reviewer confirms the same on the artwork. Neither has been built or measured
4.16	Is the released routing sound? (SR-11)	A DRC report for the released Gerber revision showing zero violations, with the isolation rule area clean on all four copper layers. The Rev B report of 2026-09-01 meets that half of this item. What it does not carry: a named layout engineer's reading of the routing, including the 169 connections the router closed at relaxed geometry – 36 below the 0.25 mm preferred conductor width, 133 at full width with a reduced gap – and a statement of which of them must change; and a written confirmation that the zoning, star-point and electrode-clearance arguments of sections 4 and 5 still hold on the routing as finally released, and not only on the Rev B routing they were written against

#	Question	Acceptance criterion
4.17	Is the channel 11 protection topology accepted as changed, and is SR-12 disposed of? (SR-12)	A written disposition of SR-12. The reviewer sees that D11.3 and C11.1 were moved from BIAS_EL to BIASOUT on 2026-09-02, that SF-1a and SF-6a of section 4.4 are closed by the move and that channel 11 is now the sixteenth ordinary channel at 36.8 µA bound A and 30.0 µA bound B (53.2 µA and 41.2 µA when this item was written, before ECO-EEG-024 was applied the same day), and either accepts the change or directs one of the other two options section 10 lists. Three things are accepted or refused with it: that C11 stays fitted at 10 nF C0G, which is what E-07 requires and what keeps the bias node terminated at radio frequency; that the new ICD-EEG-006 §2.1 assumption row on the module's bias amplifier driving 10 nF is the right home for the one question the package cannot compute, and that its consequence if untrue is functional rather than a patient-current fault; and that the minimal variant – re-point D11.3 alone, leave C11 on BIAS_EL, close SF-1a and leave SF-6a open – was available and was not taken. The reviewer may also direct the first-article contingency: depopulate C11, or the 4.7 nF C0G same-series alternate, on a written ECO

7.4 Finding classification

Class	Meaning	Disposition
Critical	Blocks Phase 2. The device may not be worn until it is closed	ECO raised, change implemented, re-review of the affected item
Major	Must be fixed before the Phase 2 build starts	ECO raised, closed before the build release
Minor	Fix at the next revision	ECO raised, tracked in ECO-EEG-016
Observation	No action required; recorded so it is not rediscovered	Recorded

Every finding gets an owner and an ECO number allocated by ECO-EEG-016. Findings SR-01 to SR-12 in this document are the programme's own opening list and are presented to the reviewer as findings, not as closed items. SR-08 is the one exception: it

is closed, and it is left in the list with its closure argument so the reviewer can disagree with it. **SR-12 is not a second exception.** The design change its analysis asked for was made on 2026-09-02 and the two fault rows it opened are closed by it, but the finding is the reviewer's and it stays open until the reviewer writes its disposition at checklist item 4.17. A programme closing its own safety finding by doing what the finding recommended is the thing this section exists to prevent.

7.5 Sign-off record

ELECTRICAL SAFETY REVIEW -- SIGN-OFF RECORD
Rev B, section 7.5

RISK-EEG-011

Reviewer
name
Registration / chartered body and
number
Declaration of independence signed (7.1) [] yes [] no
Review dates
Location

Unit reviewed: serial TI0V-B-..... hardware
rev
ATECC public-key
fingerprint
firmware image
SHA-256

Document revisions reviewed (exact revisions; these change):
RISK-EEG-011 RFQ-EEG-001 DSN-EEG-003 DSN-
EEG-002
SCH-EEG-005 ICD-EEG-006 WH-EEG-008 TST-EEG-004
QP-EEG-010 REG-EEG-012 design.py
commit
Routed Gerber revision DRC report
date
DRC violations open in that report (must be 0 -- see SR-11)

Checklist 4.1 to 4.16 completed and attached [] yes
Findings raised: Critical Major Minor
Observation

VERDICT [] Released for Phase 2
[] Released for Phase 2 with conditions (list below)
[] Not released

Conditions
.....
.....
.....

Signature
Date
Programme counter-signature
Date

Re-review triggers. Any change to a risk-control component in section 9, the M3 × 18 mm standoff included; any change to the isolation architecture, the isolator module vendor or the isolator's host connector; any change to the layer stack-up or to the reference-plane split at $x = 62$ mm; any change of cell type or supplier; any change to the E-29 acoustic ceiling or the headphone model; the Phase 2 split of the carrier for the HM-01 occipital shell, which the 150 × 130 mm Rev B board does not fit and for which no design exists; any firmware release that alters an interlock, the acoustic clamp or the lead-off excitation. QP-EEG-010 section 8 carries the same list, and ECO-EEG-016 is the mechanism that fires it.

7.6 After the review

Production and post-production information feeds back into this file: per-unit test records, refurbishment records, participant incident reports, damaged-cell returns, and support-line contacts about the device. The file is revisited when any of the following occurs – one harm of severity $\geq$ S3, two occurrences of any hazard estimated at P1 or P2, a change that fires a re-review trigger, or annually, whichever comes first.

8. Residual risks the programme accepts and states

These go into the participant information sheet and the consent form in plain words. The list below is the engineering statement; REG-EEG-012 carries the participant-facing wording, and the two must not drift apart.

#	What is accepted	What the participant is told
1	The instrument has never been built or measured, and no safety engineer has yet reviewed it. Every figure in this pack is calculated	That they are wearing an instrument built for research, that its performance has been calculated and tested but not certified, and that no one has ever worn one before them if they are in Phase 1
2	H-22: the microSD card holding their EEG, their voice and their room audio travels back through the post, unencrypted at rest	That the card is in the case, what is on it, that it carries a kit number and not their name, that it is erased after the data is taken off, and what happens if the case is lost
3	H-27: the room microphone can record their home during scripted windows	Exactly when it records, that it has a hardware mute, that the mute state is shown on screen, and that they may decline the room-audio windows
4	H-26: the controller contains a radio	That it contains a radio, that the radio is never switched on, that no network credentials exist anywhere in the device, and that every unit is swept for 60 seconds before it ships and the result is in its record

#	What is accepted	What the participant is told
5	H-14: the chin strap	That it is removable, before it is first mentioned in the fitting sequence, and that removing it costs a stability note in the record and nothing else
6	H-25: eight lights sit on their head	That the lights cannot come on by themselves at power-up, that they go dark during every recording block, and what each colour means
7	H-28: the instrument returns no result	That it cannot detect, confirm or refute anything about their experience; that no individual result exists; that the study reports groups
8	H-13: the sound level is not under their control by design, and the 100 dB SPL ceiling of E-29 is held by firmware rather than by a component	Why it is fixed, that there is a ceiling and what it is, that the red button stops the session instantly without touching the laptop, and to stop if it is uncomfortable
9	H-18, H-19: they apply an abrasive gel and a paste to their own scalp	How much, where not to put it, the contraindications, and to stop and call if skin reacts
10	H-20: the kit has been worn by other people	What is replaced between participants, what is disinfected and how, and what they never have to touch
11	H-09, H-11: there is a lithium cell in the device, and the charger cannot read the cell's temperature	Not to open the hatch, not to charge while wearing it, to charge at room temperature, and what to do if the case is dropped
12	H-29: they will be alone	The support number, that it is staffed by people who are not the research team, that they may stop or withdraw at any point without giving a reason, and that stopping costs them nothing

9. Risk-control components – no substitution

Changing any of these requires an ECO and fires a re-review under 7.5. QP-EEG-010 carries the same list as a purchasing rule; the manufacturer may not substitute them on availability.

Component	Designator	Why
ADuM4160 USB isolator module	J10	The entire isolation barrier (S-03, E-24, H-01)
Series protection resistors, 68 kΩ 0.1 % 25 ppm thin film (Vishay TNPW060368K0BEEA), ECO-EEG-024, applied 2026-09-02	R1–R16	The only current limit between a rail and the scalp (S-02, H-03). The value is the subject of SR-01 and is not a purchasing choice; a fabricator or a purchaser who substitutes 47 kΩ on availability puts S-02 back out of compliance at 53.2 μA. It also sets the E-10 band, which is ±1.0 dB at the 68 kΩ fitted and was ±0.5 dB at the 47 kΩ this row named until 2026-09-02
BAV99 dual clamp, Nexperia pinout: pin 1 AVSS, pin 2 AVDD, pin 3 the input	D1–D16	Pinout is fixed; a substitute with a different SOT-23 arrangement puts a rail on an electrode (H-03). BAT54S is not approved here: Schottky leakage across the 68 kΩ series resistor is an offset error on a 10 μV input
Filter capacitors, 10 nF C0G only , Murata GCM1885C1H103JA16D	C1–C16	X7R at +20 % fails E-10 at 100 Hz (T22); dielectric is a safety-adjacent control and an X7R part in this position is a hard reject. All sixteen are fitted, C11 included , and since 2026-09-02 all sixteen sit on the module side of their protection resistor. E-07 requires sixteen networks with C1–C16 all 10 nF C0G, so depopulating one is a breach of a mandatory requirement as well as a change to a risk-control component: it needs an ECO and it fires a re-review under 7.5, whatever a placement list says
Star-point links	R90, R91	Exactly one AGND_REF-to-DGND path and one shield path; never bridged with a wire. The rule itself is DSN-EEG-003 §3.3 rule 2
PTC and TVS on the charge input	F1, D24	Charge-path fault energy (H-09)
VBUS-present divider, 100 kΩ / 150 kΩ	R84, R85	Interlock mechanism 1 (S-01); the ratio sets whether VBUS_DET asserts at all (ECO-EEG-022)
Charger module CE net	J12.4, CHG_CE	Interlock mechanism 2 (S-01)

Component	Designator	Why
Carrier-to-plate standoff, M3 × 18 mm nylon hex, female-female	MH1–MH4	Sets the slant creepage from carrier copper to host-side module copper (S-03). A shorter standoff reopens SR-08
Session stop button, red, distinct tactile feel	SW3	The participant's only unmediated way out (H-29)
Protected 18650 with its PCM	J13 cell, HM-10 carrier	Every battery hazard (H-09 to H-11)
ATECC608B secure element	J11	Device identity; a swap detaches the calibration record from the hardware (H-23)
Two-cable harness, WH-01 and WH-02	J14, J30	Physical separation is the only control on H-05

10. Findings this pack opens

#	Finding	Class proposed	Where	Status
SR-01	Restated 2026-09-02, and open. Single-fault DC auxiliary current is 36.8 μA on the bare-resistor bound against S-02's 50 μA limit, so S-02 is met in the design. The arithmetic at the fitted 68 kΩ: bound A is 2.5 V / 68 kΩ = 36.8 μA, a 26 % margin; bound B, through the figure-12a measuring device and the thirteen parallel returns, is 2.5 V / 83.231 kΩ = 30.0 μA, 1.67$\times$. <i>Until 2026-09-02 this row read the 47 kΩ case – bound A 53.2 μA, a 6.4 % overshoot, bound B 41.2 μA – and recorded S-02 as not met. Those figures are superseded: the recommended disposition, R1–R16 to 68 kΩ with E-10 on its ± 1.0 dB branch, was taken as ECO-EEG-024 and is applied in <code>tools/design.py</code>. The finding stays open</i> for three reasons the resistor change does not touch: no figure here has been measured, no unit exists to measure, and the disposition is the reviewer's to write and not the programme's to assume. What is asked of them has changed – confirm a design that meets the limit, rather than accept one that does not. Closed on section 7.3 item 4.3, with the Phase 1 measurement made before Phase 2 is released	Critical	4.5	Open – met in the design 2026-09-02, disposition and measurement owed

#	Finding	Class proposed	Where	Status
SR-02	Both S-01 charge interlocks are firmware. No independent hardware inhibit exists	Major	6.2, H-08	Open
SR-03	No thermistor path exists from the cell to the charger. S-04's thermistor-monitored charging and E-23's 45 °C inhibit are both not met and cannot be tested	Major	6.3, H-10	Open
SR-04	E-29 now sets a 100 dB SPL ceiling, but it is enforced only by a firmware clamp on the codec volume register. There is no hardware ceiling, the artificial-ear type test TST-EEG-004 T28 has not been run, and calculated full-scale output is about 110 dB SPL	Major	H-13, 7.3 item 4.9	Open
SR-05	No biocompatibility declaration exists for any material row in REG-EEG-012 §6.2	Major	H-19	Open
SR-06	S-02 states no AC single-fault limit. Calculated SF-10 is $\approx 233 \mu\text{A}$ ($\approx 237 \mu\text{A}$ at the 47 k Ω fitted until 2026-09-02), within IEC 60601-1's 500 μA but against nothing in the RFQ	Minor	4.4	Open
SR-07	The 25-cycle reprocessing validation asked for in DSN-EEG-002 finding 8 has never been run	Major	H-21	Open

#	Finding	Class proposed	Where	Status
SR-08	The slant path across the barrier region was not guaranteed by any drawing, because the MP-01 standoff was not dimensioned	Major	5.4	Closed. The standoff is M3 × 18 mm nylon hex, four off, and MP-01 is 146.0 × 126.0 × 3.0 mm. 18.0 mm of vertical separation is more than twice the 8.0 mm required, and the 49.1 mm stack fits the 55.5 mm internal depth of POD-P1 with 6.4 mm to spare
SR-09	No ADuM4160 module vendor is fixed and no vendor board file has been obtained, so the module's own creepage is unknown	Critical	5.4	Open
SR-10	ECO-EEG-023 re-references the U7 stimulus comparator to DVDD3V3/DGND with AC-coupled inputs on a DVDD3V3/2 divider. It has not been built, measured, or reviewed for what it does to the single-star-point rule	Minor	7.3 item 4.15	Open

#	Finding	Class proposed	Where	Status
SR-11	The Rev B routing passes its design rule check: zero violations, 145 of 145 nets fully connected, one continuous plane island per net on both inner layers, and the isolation rule area clean on all four copper layers. Two things keep this finding open. The routing was produced by the programme's own tools and no human layout engineer has reviewed it, so the data is released for review under RFQ-EEG-002A and not for fabrication, and no unit exists to review. And the router relaxed 169 connections to close the board – 36 below the 0.25 mm preferred conductor width, 133 at full width with a reduced gap, all at or above the 0.20 mm minimum conductor and gap – so the section 4 and section 5 arguments are written against a routing that review may still change	Major , and a release gate ahead of the review	5.5, 7.3 item 4.16	Open, scope of RFQ-EEG-002A. The zero-violation half is met
SR-12	D11 and C11 sat on the patient side of R11, alone among the sixteen protection networks. The topology is removed. The finding is not closed. What was found: the generic loop at tools/design.py:497-500 wires every channel as src(patient) -[Rn]-dst(module) with Dn . 3 and Cn . 1 both on dst, so the 47 kΩ stands between the person and the clamp, and lines 501-503 overrode	Major	4.4, SF-1a, SF-6a, H-03, 7.3 item 4.17 (this cell cited H-08 until 2026-09-02, which was a typing error for H-03, the single-fault auxiliary-current row)	Closed in the design on 2026-09-02, open for sign-off. The three options this row listed were to move D11/C11 to BIASOUT as for the other fifteen, to add a second series element, or to accept the topology with a written justification; the

#	Finding	Class proposed	Where	Status
	channel 11 to put D11.3 and C11.1 on BIAS_EL. Whether that was chosen or slipped in while R11 was reversed, its consequences were unanalysed until SF-1a and SF-6a were written. What was done, on 2026-09-02: the override is deleted, the row is generated by the generic loop, and channel 11 is now D11.3 = C11.1 = BIASOUT with C11.2 = AGND_REF and C11 still fitted at 10 nF C0G, which E-07 requires of all sixteen. SF-1a collapses into the ordinary SF-1 – 36.8 μA bound A and 30.0 μA bound B at the 68 kΩ ECO-EEG-024 fitted the same day, where this row first read 53.2 μA and 41.2 μA – in place of 183.6 μA bound B and an unbounded bound A, and SF-6a collapses into SF-6. What is still owed: the reviewer's written disposition at checklist item 4.17, the ECO number for the change under ECO-EEG-016, and the module-qualification owner's acceptance of the new ICD-EEG-006 §2.1 assumption row on the bias amplifier driving 10 nF at its own pin. This pack does not record the finding as closed, because the programme applying the fix its own analysis recommended is not the same as the reviewer who owns the finding approving it			first was taken. The reviewer may still direct either of the others, or the minimal variant of re-pointing D11 alone, which closes SF-1a and leaves SF-6a open. Never was a routing question: no layout change altered it

#	Finding	Class proposed	Where	Status
SR-13	The DIN 42802 connector gender at J15-J17 is read from vendor documentation, not from the standard, and the package contradicts itself on it. Two independent manufacturers describe the lead-end <i>touch-proof plug</i> as ELECTRICALLY FEMALE and the instrument-end <i>input socket</i> as carrying the recessed MALE pin. WH-EEG-008 calls the same lead-end parts “1.5 mm DIN 42802 male plugs”. Neither DIN 42802 nor IEC 60601-1 has been read directly. The part itself is also unidentified – see AVL-EEG-017 §1.4.1 and ECO-EEG-036	Major	10.1 below, AVL-EEG-017 §1.4.1, WH-EEG-008 §3.1.2	Open. Opened under ECO-EEG-036; the analysis is section 10.1 and it is not closed here

10.1 SR-13 — plug or pin at J15, J16, J17

Opened under ECO-EEG-036. This section assembles the analysis for the safety reviewer. It does not settle the question and must not be read as settling it.

10.1.1 Why it was raised

docs/datasheets/README_datasheets.md established that the Stäubli part number design.py names for these three sockets appears in neither of Stäubli’s published catalogues. Looking for a replacement raised a prior question: **which end of a patient-connected DIN 42802 pair carries the male contact?** Getting that wrong puts an accessible conductive part on a lead that reaches a person.

10.1.2 What the package says, in every place it says it

Source	What it states	Gender implied
design.py J15–J17	Staubli SLB1,5-F / LB-I1,5, footprint DIN42802_1p5mm_Socket	“socket” — unstated
tools/fplib.py	docstring: “Touch-proof 1.5 mm DIN 42802 panel socket, PCB-mount version . One 1.7 mm signal pin plus two 1.5 mm mechanical retention posts”	the “1.7 mm signal pin” is the PCB solder tail , not the mating contact. Says nothing about mating gender

Source	What it states	Gender implied
Kit BOM row 36	“EMG snap leads to DIN 42802 plug , 1 m”	“plug”
WH-EEG-008 §3.1.2	electrode leads carry a “touch-proof plug ”; the harness carries a “free-hanging touch-proof socket ” it enters	socket on the instrument/harness side
WH-EEG-008 §3.1.2.1	“five identical 1.5 mm DIN 42802 male plugs ”	male on the lead
AVL-EEG-017 §1.4.1	criteria table: “One Ø1.7 mm PCB pin on the footprint centre”	again the solder tail, not the mating contact
ICD-EEG-006, IFU-EEG-014	describe the leads functionally; neither states gender	—

10.1.3 Which combinations are internally consistent

Two readings are possible, and the package contains both.

Reading A — trade usage. The instrument end is an *input socket* whose insulating body shrouds a **recessed male pin**; the lead end is a *touch-proof plug* that is **electrically female**. Under this reading: the carrier presents a pin, the footprint’s name is correct as a trade name, kit BOM row 36 is correct, WH-EEG-008 §3.1.2 is correct — **and WH-EEG-008 §3.1.2.1’s “male plugs” is wrong**, because those plugs are female.

Reading B — plain electrical usage. “Plug” means male, “socket” means female. Under this reading the carrier presents a female contact, kit BOM row 36’s male plug mates with it, §3.1.2.1 is correct — **and Stäubli’s panel-mount parts are the wrong class entirely**, since MEB1, 5-R and MLB1, 5-R present a rigid pin and would not mate with a male plug.

The two readings cannot both hold. Every document above is consistent under Reading A except one sentence; under Reading B the harness reasoning in §3.1.2 and the only catalogued Stäubli parts both fail.

10.1.4 What the evidence says, and how far it goes

It is not read from the standard. Neither DIN 42802 nor IEC 60601-1 is held by this programme, and neither has been read. **The following is asserted from vendor documentation and is marked as such.**

Source	Retrieved	What it states
The Electrode Store, “Touch-Proof Electrode Connectors”	2026-09-10	<i>“The recessed pin on the machine is the ‘male’ half of the connection, and the touch-proof safety socket on the electrode is ‘female’, although it has the appearance of a plastic pin.”</i> Also: <i>“these 1.5 mm touchproof connectors fall under the DIN 42802-1 standard (under IEC 60601-1)”</i>

Source	Retrieved	What it states
Digitimer NL844P product page	2026-09-10	"NL844P Touch-proof plugs are a 1.5mm female DIN42802 connector", mating with "the input sockets on our standalone isolated amplifiers"
Stäubli medical catalogue, doc. 11014109	2026-09-10	MEB1, 5 - R / MLB1, 5 - R: panel terminals with a "rigid, gold-plated Ø 1.5 mm pin , accepting spring-loaded Ø 1.5 mm sockets". LS1, 5 - B: "in-line, gold-plated Ø 1.5 mm sockets ... for self-assembly of connecting leads (compatible for mating with terminals such as MEB1,5-R and MLB1,5-R)"
Wikipedia, "EEG DIN connector"	2026-09-10	"touch-proof sockets around in-line rigid plugs"; "In 1989, the Deutsches Institut für Normung issued DIN 42802". Does not state which end is the instrument.

Two independent manufacturers state Reading A, and Stäubli's product shapes are consistent with it: the parts intended for a panel present pins, the parts intended for a lead are sockets. That is three lines of evidence pointing one way and none pointing the other.

It is still not the standard. A reviewer who needs this to be established rather than well-supported has to read DIN 42802-1, and this pack does not pretend otherwise.

10.1.5 Consequences of each answer

	If Reading A holds	If Reading B holds
Carrier	Presents a recessed male pin. The footprint's centre pad is the solder tail and is unaffected; no board change	Presents a female contact. Stäubli's catalogued panel parts are excluded and the candidate field narrows further
Harness	WH-EEG-008 §3.1.2.1's "male plugs" is corrected to "female"; no hardware change , a wording correction	§3.1.2's free-hanging socket at each temple would have to become a plug, changing the couplers
Leads	Kit BOM rows 36 and K2 stand as written	stand as written
Safety	The person-facing contact is female and shrouded — the intended touch-proof arrangement	A male contact somewhere on a patient-applied lead, which is what DIN 42802 exists to prevent

The asymmetry matters. Under Reading A the package needs one sentence corrected. Under Reading B it needs harness geometry rethought and has a patient-safety

concern. That is a reason to establish it properly, not a reason to assume the cheaper answer.

10.1.6 What is owed

1. A reading of **DIN 42802-1** confirming or refusing Reading A. Until then the evidence in 10.1.4 stays marked asserted.
2. A decision on WH-EEG-008 §3.1.2.1's "male plugs" — a wording correction under Reading A, a design question under Reading B. **Not corrected here**, because correcting it presumes the answer.
3. It interacts with the open part question (AVL-EEG-017 §1.4.1, ECO-EEG-036): a sample submitted for approval settles gender by inspection, and the QP-EEG-010 §11.2 criteria table should gain a **gender row** so that it does.

Nothing in the design, the harness or the BOM has been changed by this section.

11. What this document does not do

It does not make the device safe. It makes the device's hazards legible, states the arithmetic that supports each control, and hands a named external engineer a defined thing to sign. Twelve findings are raised. SR-08 is closed; SR-12's design change was made on 2026-09-02, which closes the two fault rows it opened and leaves the finding itself open for the reviewer's disposition; ten others are open, and two of those are Critical. **Restated 2026-09-02:** this paragraph said that three mandatory requirements – S-02, S-04 and E-23 – were recorded here as not met. **Two are:** S-04 and E-23's 45 °C-inhibit half, both for want of a thermistor path that does not exist (SR-03). **S-02 is met in the design** at 36.8 µA since ECO-EEG-024 took R1–R16 to 68 kΩ, and what is owed on it is the reviewer's disposition and a measurement, not a design change (SR-01). This pack asks the reviewer to decide what happens about all three rather than presenting any of them as closed. The Rev B routing now passes every design rule with zero violations, every net one connected copper island and both inner planes continuous under the analogue zone – the three conditions of the ECO-EEG-016 section 3 gate – so the fabrication data is released for review under RFQ-EEG-002A; it is still not released for fabrication, because no layout engineer has reviewed it and 169 connections were closed at relaxed geometry (SR-11). Nothing here has been built or measured. No unit goes on a head until section 7.5 is signed.

Rev	Date	Change
A	2026-09-01	First issue. Written against EEG-CAR-01 Rev B and design.py of 2026-09-01. Closes the risk-management-file, patient-aux-current-budget, isolation-creepage-verification and safety-review-gate-pack gaps; opens SR-01 to SR-09
B	2026-09-01	Carrier corrected to 150.0 × 130.0 mm and to a four-layer stack with reference planes on L2 and L3, with the barrier keep-out and its verification extended to all four layers and a new fault case SF-13; SR-08 closed by the M3 × 18 mm standoff and the 146 × 126 × 3 mm MP-01 plate; SR-01's recommended fix restated as 68 kΩ / ECO-EEG-024 with E-10 widened to ±1.0 dB, and S-02 kept open as a Phase 1 decision; E-29's 100 dB

Rev	Date	Change
		SPL acoustic ceiling adopted and SR-04 rewritten against it; S-04 and E-23 stated as not met beside the requirement; HM-07/HM-08/HM-10 naming corrected; R85 corrected to 150 kΩ; SR-10 opened for ECO-EEG-023. Corrections within Rev B, after the second cross-document audit of 2026-09-01: findings 1.7, 1.8 and 1.9 are closed. Section 5.5 rewritten from the real DRC report: the isolation rule area is recorded as clean on all four copper layers because the report says so, and the DRC state is stated as a release gate, with SR-11 and checklist item 4.16 opened for it. E-10's two states, ± 0.5 dB with the 47 kΩ fitted and ± 1.0 dB only if ECO-EEG-024 is taken, are now stated together everywhere E-10 appears. SR-01 keeps its arithmetic and stays open. The maximum-acoustic-output type test is cited as TST-EEG-004 T28 of TST section 14, and the creepage type test as TST-EEG-004 section 14. Corrections within Rev B, after the verification review of package v2.1 of 1 September 2026: H-16's packaging control is restated against the CASE-00 Rev C seven-layer insert, with the drop type test noted as having no cut insert to run against yet, and H-24's lead-off references are the E96 values 4k99 / 10k0 / 49k9 with the calibration described as TST-EEG-004 T10 actually performs it. Corrections within Rev B, after the routing closed on 2026-09-01: the DRC report now records zero violations with 145 of 145 nets connected and both inner planes continuous, so the gate of ECO-EEG-016 section 3 is met and the fabrication data is released for review under RFQ-EEG-002A; it remains not released for fabrication, because no human layout engineer has reviewed the routing. Section 5.5, section 7, receipt checklist item 7, checklist item 4.16, SR-11 and section 11 are restated on that basis, the DRC evidence item is marked satisfied, the 169 connections the router closed at relaxed geometry are stated wherever routing quality is discussed, and the electrode-clearance breaches the earlier issue recorded are closed and removed. S-02, S-04 and E-23 are untouched by the routing and stay not met. Corrections within Rev B, after the independent review of package v2.2 of 2026-09-02: H-16 is restated against what mech/ now holds – the seven CASE-00 Rev C layer cut files are shipped and the Rev B pair is deleted, and the honest half of the row survives because they are not released for cutting and no insert has been cut; section 6.1 and receipt-checklist item 9 mark the superseded sections of DSN-EEG-002 Rev E, which has not been re-issued; and the two

Rev	Date	Change
		summary restatements of the ECO-EEG-016 section 3 gate now name all three of its conditions, as section 5.5 already did. Corrections within Rev B, after the design changes of 2 September 2026: channel 11's protection network is generated by the same loop as the other fifteen, so SF-1a and SF-6a of section 4.4 are restated as closed by that change with the ordinary SF-1 figures and the 183.6 μA is withdrawn, section 4.1 states the topology, section 4.4 gains the two things the change does not close, SR-12 is restated as closed in the design and open for sign-off, checklist item 4.17 is added for its disposition, section 9's C1–C16 row states that all sixteen are fitted, and section 11's finding count is corrected from eleven to twelve; SF-9 and H-05 are restated against the HM-04 body's two separated pockets with 1.60 mm of PA12 between them, as a material reduction that is explicitly not an elimination and that has been printed on nothing; H-17 records that the contact force its control rests on is still an unspecified spring and that HM-04's deepened seat now leaves 3.50 mm of free height for one; and H-25's "not yet coded" is superseded, E-27's bicolour phase driver having been written on that date. Corrections within Rev B, after ECO-EEG-024 was applied on 2026-09-02. R1–R16 are 68 kΩ, not 47 kΩ, so every single-fault figure in section 4 is restated: bound A 36.8 μA and bound B 30.0 μA against S-02's 50 μA, bound B's loop 83.231 kΩ, SF-1/SF-2/SF-3 passing on both bounds, SF-8 73.5 μA, SF-10 $\approx 233 \mu\text{A}$, the section 4.5 conclusion table closing on both bounds, and the "what is actually built" paragraph corrected on both of its errors – no prototype has been fabricated, and ECO-EEG-024 is applied rather than open. S-02 is met in the design and SR-01 stays open, because the finding is the safety reviewer's, nothing is measured and no unit exists; SR-01, checklist item 4.3, H-03, H-06, section 9's resistor and BAV99 rows, section 11's "three mandatory requirements" count and H-24's lead-off and offset figures are restated on that basis. E-10 sits on its ± 1.0 dB branch, which the requirement already states. Also corrected on the same date: the re-route for the channel 11 change is complete – the 2026-09-02 report gives 145 of 145 nets, VIOLATIONS: 0, 3 745 segments and 552 vias, so receipt item 7 is satisfied against that report and not the 2026-09-01 one – and the expectation that the relaxed "BIASOUT J2.10 -> R11.1" connection would retire with the change is withdrawn: it is still listed at 0.20 mm track and 0.35 mm gap. H-25 additionally records that the light alternation quantises to the FreeRTOS tick at about 250 Hz rather than exactly 240